

TRIGONOMETRIC FUNCTIONS

5.1 INTRODUCTION

In the present chapter, we will first introduce trigonometric ratios which are also known as trigonometric functions and then the identities involving them.

5.2 TRIGONOMETRIC RATIOS OR FUNCTIONS

Consider an angle $\theta = \angle XO A$ as shown in Fig. 5.1. Let P be any point other than O on its terminal side OA and let PM be perpendicular from P on x -axis. Let length $OP = r$, $OM = x$ and $MP = y$. We take the length $OP = r$ always positive while x and y can be positive or negative depending upon the position of the terminal side OA of $\angle XO A$.

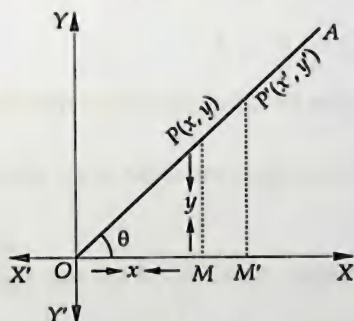


Fig. 5.1

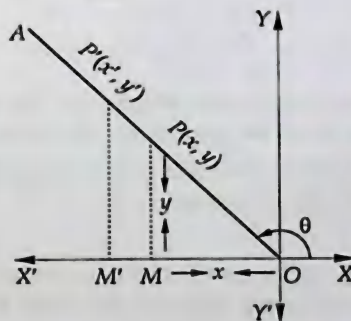


Fig. 5.2

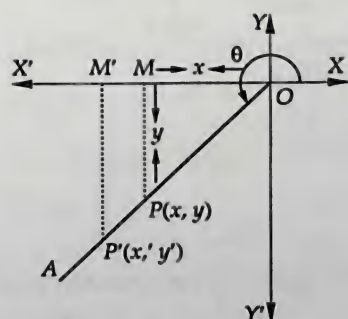


Fig. 5.3

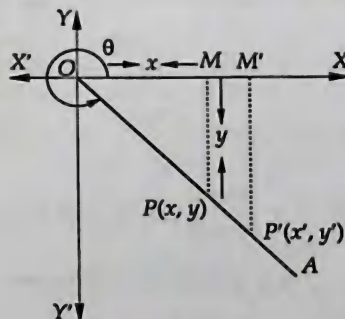


Fig. 5.4

In the right angled triangle OMP , we have

Base = $OM = x$, Perpendicular = $PM = y$, and Hypotenues = $OP = r$

We define the following trigonometric ratios which are also known as trgonometric functions.

Sine $\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{y}{r}$, and is written as $\sin \theta$

Cosine $\theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{x}{r}$, and is written as $\cos \theta$

$$\begin{aligned}\text{Tangent } \theta &= \frac{\text{Perpendicular}}{\text{Base}} = \frac{y}{x}, \text{ and is written as } \tan \theta \\ \text{Cosecant } \theta &= \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{r}{y}, \text{ and is written as } \operatorname{cosec} \theta \\ \text{Secant } \theta &= \frac{\text{Hypotenuse}}{\text{Base}} = \frac{r}{x}, \text{ and is written as } \sec \theta \\ \text{Cotangent } \theta &= \frac{\text{Base}}{\text{Perpendicular}} = \frac{x}{y}, \text{ and is written as } \cot \theta\end{aligned}$$

NOTE 1 The triangle OMP is known as the triangle of reference.

NOTE 2 It should be noted that $\sin \theta$ does not mean the product of $\sin$ and θ . The $\sin \theta$ is correctly read as sine of angle θ .

These functions depend only on the value of the angle θ and not on the position of the point P chosen on the terminal side of the angle θ as proved in the following theorem.

THEOREM The trigonometric ratios are same for the same angle.

PROOF Let $P'(x', y')$ be any point other than $P(x, y)$ on the terminal side OA with $OP' = r'$. Let $P' M'$ be perpendicular on x -axis. Clearly, triangles OMP and $OM' P'$ are similar. Therefore,

$$\frac{y}{r} = \frac{y'}{r'}, \frac{x}{r} = \frac{x'}{r'} \quad \text{and} \quad \frac{y}{x} = \frac{y'}{x'}$$

$$\Rightarrow \sin \theta = \frac{y}{r} = \frac{y'}{r'}, \quad \cos \theta = \frac{x}{r} = \frac{x'}{r'} \quad \text{and} \quad \tan \theta = \frac{y}{x} = \frac{y'}{x'}$$

Thus, sine, cosine and tangent are the same whatever point be taken on the terminal side OA . Similarly, it can be proved for the other ratios.

Hence, the trigonometric ratios or trigonometric functions are independent of the choice of the size of triangle of reference.

Q.E.D.

REMARK 1 If the terminal side coincides with x -axis, then $\operatorname{cosec} \theta$ and $\cot \theta$ are not defined. If it coincides with y -axis, then $\sec \theta$ and $\tan \theta$ are not defined.

REMARK 2 The following relations are obvious from the definitions of trigonometric ratios :

$$\begin{aligned}\text{(i)} \quad \sin \theta \times \operatorname{cosec} \theta &= 1 \Rightarrow \sin \theta = \frac{1}{\operatorname{cosec} \theta} \quad \text{and} \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta} \\ \text{(ii)} \quad \cos \theta \times \sec \theta &= 1 \Rightarrow \cos \theta = \frac{1}{\sec \theta} \quad \text{and} \quad \sec \theta = \frac{1}{\cos \theta} \\ \text{(iii)} \quad \tan \theta \times \cot \theta &= 1 \Rightarrow \cot \theta = \frac{1}{\tan \theta} \quad \text{and} \quad \tan \theta = \frac{1}{\cot \theta} \\ \text{(iv)} \quad \tan \theta &= \frac{\sin \theta}{\cos \theta} \quad \text{and} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}\end{aligned}$$

REMARK 3 The trigonometric ratios may be positive or negative depending upon x and/or y as discussed in section 5.4.

5.3 TRIGONOMETRIC IDENTITIES

IDENTITY An equation involving trigonometric functions which is true for all those angles for which the functions are defined is called a trigonometric identity.

For example, $\sec \theta = \frac{1}{\cos \theta}$, $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$ etc. are trigonometric identities as they hold for all those values of θ except those values for which $\sec \theta$ and $\operatorname{cosec} \theta$ are not defined.

But, $\sin \theta = \cos \theta$ is a trigonometric equation not a trigonometric identity because it does not hold for all values of θ .

5.3.1 FUNDAMENTAL TRIGONOMETRIC IDENTITIES

In this subsection, we shall state and prove some fundamental trigonometrical identities.

THEOREM Prove that:

$$(i) \sin \theta = \frac{1}{\operatorname{cosec} \theta} \text{ or, } \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad (ii) \cos \theta = \frac{1}{\sec \theta} \text{ or, } \sec \theta = \frac{1}{\cos \theta}$$

$$(iii) \cot \theta = \frac{1}{\tan \theta} \text{ or, } \tan \theta = \frac{1}{\cot \theta} \quad (iv) \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$(v) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$(vi) \sin^2 \theta + \cos^2 \theta = 1$$

$$(vii) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(viii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

PROOF Let a revolving ray start from OX and revolve into the position OP to trace out any angle θ in any of the four quadrants. From P draw PM perpendicular to x -axis. In the right angled triangle OMP , we have

$$OP^2 = OM^2 + PM^2,$$

$$\sin \theta = \frac{PM}{OP}, \cos \theta = \frac{OM}{OP}, \tan \theta = \frac{PM}{OM}, \operatorname{cosec} = \frac{OP}{PM}, \sec \theta = \frac{OP}{OM} \text{ and, } \cot \theta = \frac{OM}{PM}.$$

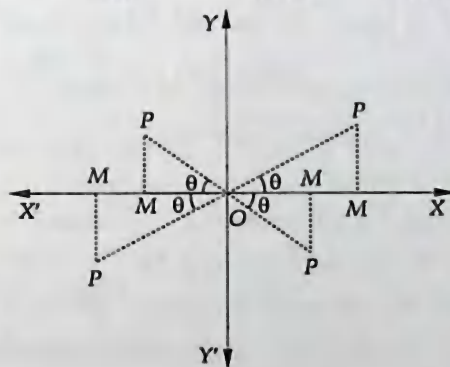


Fig. 5.5

Clearly, identities in (i) to (v) are trivial. So, let us prove identity (vi).

$$(vi) \sin^2 \theta + \cos^2 \theta = \left(\frac{PM}{OP}\right)^2 + \left(\frac{OM}{OP}\right)^2 = \frac{PM^2 + OM^2}{OP^2} = \frac{OP^2}{OP^2} = 1$$

$$\text{Hence, } \sin^2 \theta + \cos^2 \theta = 1$$

(vii) We have,

$$1 + \tan^2 \theta = 1 + \left(\frac{PM}{OM}\right)^2 = 1 + \frac{PM^2}{OM^2} = \frac{OM^2 + PM^2}{OM^2} = \frac{OP^2}{OM^2} = \left(\frac{OP}{OM}\right)^2 = \sec^2 \theta$$

$$\text{Hence, } 1 + \tan^2 \theta = \sec^2 \theta.$$

(viii) We have,

$$1 + \cot^2 \theta = 1 + \left(\frac{OM}{PM}\right)^2 = 1 + \frac{OM^2}{PM^2} = \frac{PM^2 + OM^2}{PM^2} = \frac{OP^2}{PM^2} = \left(\frac{OP}{PM}\right)^2 = \operatorname{cosec}^2 \theta$$

$$\text{Hence, } 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Q.E.D.

NOTE It should be noted that $(\sin \theta)^2$ is written as $\sin^2 \theta$, $(\cos \theta)^2$ is written as $\cos^2 \theta$ etc.

We shall now discuss more identities involving the trigonometrical functions in the following examples.

ILLUSTRATIVE EXAMPLES**LEVEL-1****EXAMPLE 1** Prove the following identities:

(i) $\sin^8 \theta - \cos^8 \theta = (\sin^2 \theta - \cos^2 \theta)(1 - 2 \sin^2 \theta \cos^2 \theta)$

(ii) $\cot^4 \theta + \cot^2 \theta = \operatorname{cosec}^4 \theta - \operatorname{cosec}^2 \theta$

(iii) $2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta = \cot^4 \theta - \tan^4 \theta$

(iv) $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \tan^2 \theta + \cot^2 \theta + 7$

SOLUTION (i) We have,

$$\begin{aligned}
 \text{LHS} &= (\sin^8 \theta - \cos^8 \theta) = (\sin^4 \theta)^2 - (\cos^4 \theta)^2 \\
 &= (\sin^4 \theta - \cos^4 \theta)(\sin^4 \theta + \cos^4 \theta) \\
 &= (\sin^2 \theta - \cos^2 \theta)(\sin^2 \theta + \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta) \\
 &= (\sin^2 \theta - \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta) \\
 &= (\sin^2 \theta - \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta - 2 \sin^2 \theta \cos^2 \theta) \\
 &= (\sin^2 \theta - \cos^2 \theta) \left\{ (\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta \right\} \\
 &= (\sin^2 \theta - \cos^2 \theta)(1 - 2 \sin^2 \theta \cos^2 \theta) = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) LHS} &= \cot^4 \theta + \cot^2 \theta = (\cot^2 \theta)^2 + \cot^2 \theta \\
 &= (\operatorname{cosec}^2 \theta - 1)^2 + (\operatorname{cosec}^2 \theta - 1) \quad [\because 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta] \\
 &= \operatorname{cosec}^4 \theta - 2 \operatorname{cosec}^2 \theta + 1 + \operatorname{cosec}^2 \theta - 1 = \operatorname{cosec}^4 \theta - \operatorname{cosec}^2 \theta = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) LHS} &= 2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta \\
 &= 2 \sec^2 \theta - (\sec^2 \theta)^2 - 2 \operatorname{cosec}^2 \theta + (\operatorname{cosec}^2 \theta)^2 \\
 &= 2(1 + \tan^2 \theta) - (1 + \tan^2 \theta)^2 - 2(1 + \cot^2 \theta) + (\cot^2 \theta + 1)^2 \\
 &= 2 + 2 \tan^2 \theta - (1 + \tan^4 \theta + 2 \tan^2 \theta) - 2 - 2 \cot^2 \theta + (\cot^4 \theta - 2 \cot^2 \theta + 1) \\
 &= \cot^4 \theta - \tan^4 \theta = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) LHS} &= (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 \\
 &= \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 \sin \theta \operatorname{cosec} \theta + \cos^2 \theta + \sec^2 \theta + 2 \cos \theta \sec \theta \\
 &= (\sin^2 \theta + \cos^2 \theta) + (\operatorname{cosec}^2 \theta + \sec^2 \theta) + 2 + 2 \\
 &= 1 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) + 4 = \tan^2 \theta + \cot^2 \theta + 7 = \text{RHS}
 \end{aligned}$$

EXAMPLE 2 Prove the following identities:

(i) $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2$

(ii) $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$

[NCERT EXEMPLAR]**SOLUTION** (i) We have,

$$\begin{aligned}
 \text{LHS} &= (1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) \\
 &= \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right) \\
 &= \frac{(\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}{\sin \theta \cos \theta}
 \end{aligned}$$

$$= \frac{(\sin \theta + \cos \theta)^2 - 1}{\sin \theta \cos \theta} = \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} = \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} = 2 = \text{RHS}$$

$$\begin{aligned} \text{(ii) LHS} &= \frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{(\tan \theta + \sec \theta) - (\sec^2 \theta - \tan^2 \theta)}{\tan \theta - \sec \theta + 1} \quad [\because \sec^2 \theta - \tan^2 \theta = 1] \\ &= \frac{(\sec \theta + \tan \theta) [1 - (\sec \theta - \tan \theta)]}{\tan \theta - \sec \theta + 1} = \frac{(\sec \theta + \tan \theta) (\tan \theta - \sec \theta + 1)}{\tan \theta - \sec \theta + 1} \\ &= \sec \theta + \tan \theta = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \frac{1 + \sin \theta}{\cos \theta} = \text{RHS} \end{aligned}$$

EXAMPLE 3 If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$, show that $m^2 - n^2 = 4\sqrt{mn}$.

SOLUTION We have,

[NCERT EXEMPLAR]

$$\text{LHS} = m^2 - n^2 = (\tan \theta + \sin \theta)^2 - (\tan \theta - \sin \theta)^2 = 4 \tan \theta \sin \theta \quad \dots \text{(i)}$$

$$\text{RHS} = 4\sqrt{mn} = 4\sqrt{(\tan \theta + \sin \theta)(\tan \theta - \sin \theta)} = 4\sqrt{\tan^2 \theta - \sin^2 \theta}$$

$$\Rightarrow \text{RHS} = 4\sqrt{\frac{\sin^2 \theta}{\cos^2 \theta} - \sin^2 \theta} = 4\sqrt{\frac{\sin^2 \theta - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta}} = 4\sqrt{\frac{\sin^2 \theta (1 - \cos^2 \theta)}{\cos^2 \theta}}$$

$$\Rightarrow \text{RHS} = 4\sqrt{\frac{\sin^4 \theta}{\cos^2 \theta}} = 4\frac{\sin^2 \theta}{\cos \theta} = 4 \tan \theta \sin \theta \quad \dots \text{(ii)}$$

From (i) and (ii), we obtain that LHS = RHS.

EXAMPLE 4 If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, prove that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$.

SOLUTION We have,

$$\cos \theta + \sin \theta = \sqrt{2} \cos \theta$$

$$\Rightarrow (\cos \theta + \sin \theta)^2 = (\sqrt{2} \cos \theta)^2$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta = 2 \cos^2 \theta$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow (\cos \theta + \sin \theta)(\cos \theta - \sin \theta) = 2 \sin \theta \cos \theta$$

$$\Rightarrow \cos \theta - \sin \theta = \frac{2 \sin \theta \cos \theta}{\cos \theta + \sin \theta}$$

$$\Rightarrow \cos \theta - \sin \theta = \frac{2 \sin \theta \cos \theta}{\sqrt{2} \cos \theta} \quad [\because \cos \theta + \sin \theta = \sqrt{2} \cos \theta]$$

$$\Rightarrow \cos \theta - \sin \theta = \sqrt{2} \sin \theta$$

ALITER We know that

$$(\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 = 2$$

$$\Rightarrow (\sqrt{2} \cos \theta)^2 + (\cos \theta - \sin \theta)^2 = 2 \quad [\because \cos \theta + \sin \theta = \sqrt{2} \cos \theta]$$

$$\Rightarrow (\cos \theta - \sin \theta)^2 = 2 - 2 \cos^2 \theta$$

$$\Rightarrow (\cos \theta - \sin \theta)^2 = 2 \sin^2 \theta \Rightarrow \cos \theta - \sin \theta = \sqrt{2} \sin \theta.$$

EXAMPLE 5 If $a \cos \theta + b \sin \theta = x$ and $a \sin \theta - b \cos \theta = y$, prove that $a^2 + b^2 = x^2 + y^2$.

[NCERT EXEMPLAR]

SOLUTION We have,

$$x = a \cos \theta + b \sin \theta \text{ and } y = a \sin \theta - b \cos \theta.$$

$$\begin{aligned}
 \therefore x^2 + y^2 &= (a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2 \\
 &= (a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \sin \theta \cos \theta) + (a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab \sin \theta \cos \theta) \\
 &= a^2 (\cos^2 \theta + \sin^2 \theta) + b^2 (\sin^2 \theta + \cos^2 \theta) = a^2 + b^2
 \end{aligned}$$

EXAMPLE 6 If $a \cos \theta - b \sin \theta = c$, show that $a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}$.

SOLUTION Clearly,

$$\begin{aligned}
 &(a \cos \theta - b \sin \theta)^2 + (a \sin \theta + b \cos \theta)^2 \\
 &= a^2 (\cos^2 \theta + \sin^2 \theta) + b^2 (\sin^2 \theta + \cos^2 \theta) - 2ab \sin \theta \cos \theta + 2ab \sin \theta \cos \theta \\
 &= a^2 + b^2
 \end{aligned}$$

$$\therefore (a \sin \theta + b \cos \theta)^2 = a^2 + b^2 - (a \cos \theta - b \sin \theta)^2$$

$$\Rightarrow (a \sin \theta + b \cos \theta)^2 = a^2 + b^2 - c^2$$

$$[\because a \cos \theta - b \sin \theta = c]$$

$$\Rightarrow a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}$$

EXAMPLE 7 If $\sec \theta + \tan \theta = p$, obtain the values of $\sec \theta$, $\tan \theta$ and $\sin \theta$ in terms of p .

SOLUTION We know that: $\sec^2 \theta - \tan^2 \theta = 1$.

$$\therefore (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1 \Rightarrow p(\sec \theta - \tan \theta) = 1 \Rightarrow \sec \theta - \tan \theta = \frac{1}{p}$$

$$\text{Now, } \sec \theta + \tan \theta = p \text{ and, } \sec \theta - \tan \theta = \frac{1}{p}$$

$$\Rightarrow (\sec \theta + \tan \theta) + (\sec \theta - \tan \theta) = p + \frac{1}{p}$$

$$\Rightarrow 2 \sec \theta = p + \frac{1}{p} \Rightarrow \sec \theta = \frac{p^2 + 1}{2p} \quad \dots(i)$$

$$\text{Again, } \sec \theta + \tan \theta = p \text{ and } \sec \theta - \tan \theta = \frac{1}{p}$$

$$\Rightarrow (\sec \theta + \tan \theta) - (\sec \theta - \tan \theta) = p - \frac{1}{p}$$

$$\Rightarrow 2 \tan \theta = p - \frac{1}{p} \Rightarrow \tan \theta = \frac{p^2 - 1}{2p} \quad \dots(ii)$$

$$\text{Dividing (ii) by (i), we get: } \sin \theta = \frac{p^2 - 1}{p^2 + 1}$$

EXAMPLE 8 Prove that: $2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta = \frac{1 - \tan^8 \theta}{\tan^4 \theta}$.

SOLUTION We have,

$$\begin{aligned}
 &2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta \\
 &= 2(1 + \tan^2 \theta) - (1 + \tan^2 \theta)^2 - 2(1 + \cot^2 \theta) + (1 + \cot^2 \theta)^2 \\
 &= 2(1 + \tan^2 \theta - 1 - \cot^2 \theta) + (1 + 2 \cot^2 \theta + \cot^4 \theta) - (1 + 2 \tan^2 \theta + \tan^4 \theta) \\
 &= 2(\tan^2 \theta - \cot^2 \theta) + (2 \cot^2 \theta - 2 \tan^2 \theta) + \cot^4 \theta - \tan^4 \theta \\
 &= \cot^4 \theta - \tan^4 \theta = \frac{1}{\tan^4 \theta} - \tan^4 \theta = \frac{1 - \tan^8 \theta}{\tan^4 \theta}
 \end{aligned}$$

EXAMPLE 9 Prove that: $3(\sin \theta - \cos \theta)^4 + 6(\sin \theta + \cos \theta)^2 + 4(\sin^6 \theta + \cos^6 \theta) - 13 = 0$.

SOLUTION We have,

$$\begin{aligned}
 & 3(\sin \theta - \cos \theta)^4 + 6(\sin \theta + \cos \theta)^2 + 4(\sin^6 \theta + \cos^6 \theta) - 13 \\
 &= 3 \left\{ (\sin \theta - \cos \theta)^2 \right\}^2 + 6(\sin \theta + \cos \theta)^2 \\
 &\quad + 4 \left\{ (\sin^2 \theta + \cos^2 \theta)^3 - 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta) \right\} - 13 \\
 &= 3(1 - 2 \sin \theta \cos \theta)^2 + 6(1 + 2 \sin \theta \cos \theta) + 4(1 - 3 \sin^2 \theta \cos^2 \theta) - 13 \\
 &= 3(1 - 4 \sin \theta \cos \theta + 4 \sin^2 \theta \cos^2 \theta) + 6(1 + 2 \sin \theta \cos \theta) + 4(1 - 3 \sin^2 \theta \cos^2 \theta) - 13 \\
 &= 3 + 6 + 4 - 13 = 0
 \end{aligned}$$

EXAMPLE 10 Given that: $(1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma) = (1 - \cos \alpha)(1 - \cos \beta)(1 - \cos \gamma)$.

Show that one of the values of each member of this equality is $\sin \alpha \sin \beta \sin \gamma$.

SOLUTION We have,

$$(1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma) = (1 - \cos \alpha)(1 - \cos \beta)(1 - \cos \gamma)$$

Multiplying both sides by $(1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma)$, we get

$$(1 + \cos \alpha)^2 (1 + \cos \beta)^2 (1 + \cos \gamma)^2 = (1 - \cos \alpha)(1 - \cos \beta)(1 - \cos \gamma) (1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma)$$

$$\Rightarrow (1 + \cos \alpha)^2 (1 + \cos \beta)^2 (1 + \cos \gamma)^2 = (1 - \cos^2 \alpha)(1 - \cos^2 \beta)(1 - \cos^2 \gamma)$$

$$\Rightarrow (1 + \cos \alpha)^2 (1 + \cos \beta)^2 (1 + \cos \gamma)^2 = \sin^2 \alpha \sin^2 \beta \sin^2 \gamma$$

$$\Rightarrow (1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma) = \pm \sin \alpha \sin \beta \sin \gamma$$

Hence, one of the values of $(1 + \cos \alpha)(1 + \cos \beta)(1 + \cos \gamma)$ is $\sin \alpha \sin \beta \sin \gamma$.

Similarly, by multiplying both sides by $(1 - \cos \alpha)(1 - \cos \beta)(1 - \cos \gamma)$, we find that one of the values of $(1 - \cos \alpha)(1 - \cos \beta)(1 - \cos \gamma)$ is also $\sin \alpha \sin \beta \sin \gamma$.

LEVEL-2

EXAMPLE 11 Prove that: $\sec^2 \theta + \operatorname{cosec}^2 \theta \geq 4$.

SOLUTION We have,

$$\begin{aligned}
 \sec^2 \theta + \operatorname{cosec}^2 \theta &= (1 + \tan^2 \theta) + 1 + (\cot^2 \theta) = 2 + \tan^2 \theta + \cot^2 \theta \\
 &= 2 + \tan^2 \theta + \cot^2 \theta - 2 \tan \theta \cot \theta + 2 \tan \theta \cot \theta \\
 &= 2 + (\tan \theta - \cot \theta)^2 + 2 \\
 &= 4 + (\tan \theta - \cot \theta)^2 \geq 4 \quad [\because (\tan \theta - \cot \theta)^2 \geq 0]
 \end{aligned}$$

EXAMPLE 12 If $10 \sin^4 \alpha + 15 \cos^4 \alpha = 6$, find the value of $27 \operatorname{cosec}^6 \alpha + 8 \sec^6 \alpha$.

SOLUTION We have,

$$10 \sin^4 \alpha + 15 \cos^4 \alpha = 6$$

$$\Rightarrow 10 \sin^4 \alpha + 15 \cos^4 \alpha = 6(\sin^2 \alpha + \cos^2 \alpha)^2$$

$$\Rightarrow 10 \tan^4 \alpha + 15 = 6(\tan^2 \alpha + 1)^2 \quad [\text{Dividing both sides by } \cos^4 \alpha]$$

$$\Rightarrow (2 \tan^2 \alpha - 3)^2 = 0 \Rightarrow \tan^2 \alpha = \frac{3}{2}$$

$$\begin{aligned}
 \therefore 27 \operatorname{cosec}^6 \alpha + 8 \sec^6 \alpha &= 27(1 + \cot^2 \alpha)^3 + 8(1 + \tan^2 \alpha)^3 \\
 &= 27 \left(1 + \frac{2}{3}\right)^3 + 8 \left(1 + \frac{3}{2}\right)^3 = 27 \times \frac{125}{27} + 8 \times \frac{125}{8} = 250.
 \end{aligned}$$

EXAMPLE 13 If $\frac{\sin A}{\sin B} = p$ and $\frac{\cos A}{\cos B} = q$, find $\tan A$ and $\tan B$.

SOLUTION We have,

$$\frac{\sin A}{\sin B} = p \text{ and } \frac{\cos A}{\cos B} = q$$

$$\Rightarrow \frac{\sin A}{\sin B} \cdot \frac{\cos B}{\cos A} = \frac{p}{q}$$

$$\Rightarrow \frac{\tan A}{\tan B} = \frac{p}{q} \Rightarrow \frac{\tan A}{p} = \frac{\tan B}{q} = \lambda (\text{say}) \Rightarrow \tan A = p\lambda \text{ and } \tan B = q\lambda \quad \dots(i)$$

Now, $\sin A = p \sin B$

$$\Rightarrow \frac{\tan A}{\sqrt{1 + \tan^2 A}} = p \frac{\tan B}{\sqrt{1 + \tan^2 B}}$$

$$\Rightarrow \frac{p\lambda}{\sqrt{1 + p^2 \lambda^2}} = p \frac{q\lambda}{\sqrt{1 + q^2 \lambda^2}}$$

$$\Rightarrow p^2 (1 + q^2 \lambda^2) = p^2 q^2 (1 + p^2 \lambda^2)$$

$$\Rightarrow \lambda^2 (q^2 - p^2 q^2) = q^2 - 1$$

$$\Rightarrow \lambda^2 = \frac{q^2 - 1}{q^2 (1 - p^2)} \Rightarrow \lambda = \pm \frac{1}{q} \sqrt{\frac{q^2 - 1}{1 - p^2}}$$

$$\therefore \tan A = \pm \frac{p}{q} \sqrt{\frac{q^2 - 1}{1 - p^2}} \text{ and, } \tan B = \pm \sqrt{\frac{q^2 - 1}{1 - p^2}}$$

[Using (i)]

EXAMPLE 14 If $\tan^2 \theta = 1 - a^2$, prove that $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = (2 - a^2)^{3/2}$.

Also, find the values of a for which the above result holds true.

SOLUTION We have,

$$\begin{aligned} \sec \theta + \tan^3 \theta \operatorname{cosec} \theta &= \sec \theta \left\{ 1 + \tan^3 \theta \frac{\operatorname{cosec} \theta}{\sec \theta} \right\} \\ &= \sqrt{1 + \tan^2 \theta} \left\{ 1 + \tan^3 \theta \times \cot \theta \right\} \\ &= (1 + \tan^2 \theta)^{3/2} \\ &= (1 + 1 - a^2)^{3/2} = (2 - a^2)^{3/2} \end{aligned}$$

$$[\because \tan^2 \theta = 1 - a^2]$$

Now,

$$\tan^2 \theta \geq 0 \text{ for all } \theta \Rightarrow 1 - a^2 \geq 0 \Rightarrow a^2 - 1 \leq 0 \Rightarrow -1 \leq a \leq 1 \quad \dots(i)$$

Since LHS of $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = (2 - a^2)^{3/2}$ is real for all $\theta \in \mathbb{R}$. So, RHS must also be real.

$$\therefore 2 - a^2 \geq 0 \Rightarrow a^2 - 2 \leq 0 \Rightarrow -\sqrt{2} \leq a \leq \sqrt{2} \quad \dots(ii)$$

From (i) and (ii), we find that the given relation holds true for all $a \in [-1, 1]$.

EXAMPLE 15 If $a \cos^3 \theta + 3a \cos \theta \sin^2 \theta = m$ and $a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = n$, then prove that: $(m+n)^{2/3} + (m-n)^{2/3} = 2a^{2/3}$.

SOLUTION We have,

$$a \cos^3 \theta + 3a \cos \theta \sin^2 \theta = m \text{ and } a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = n$$

$$\Rightarrow a \cos^3 \theta + 3a \cos \theta \sin^2 \theta + a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = m + n$$

$$\text{and, } a \cos^3 \theta + 3a \cos \theta \sin^2 \theta - a \sin^3 \theta - 3a \cos^2 \theta \sin \theta = m - n$$

$$\Rightarrow a(\cos \theta + \sin \theta)^3 = m + n \quad \text{and, } a(\cos \theta - \sin \theta)^3 = m - n$$

$$\Rightarrow \cos \theta + \sin \theta = \left(\frac{m+n}{a} \right)^{1/3} \quad \text{and, } \cos \theta - \sin \theta = \left(\frac{m-n}{a} \right)^{1/3}$$

$$\Rightarrow (\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 = \left(\frac{m+n}{a} \right)^{2/3} + \left(\frac{m-n}{a} \right)^{2/3}$$

$$\Rightarrow 2(\cos^2 \theta + \sin^2 \theta) = \frac{(m+n)^{2/3}}{a^{2/3}} + \frac{(m-n)^{2/3}}{a^{2/3}}$$

$$\Rightarrow (m+n)^{2/3} + (m-n)^{2/3} = 2a^{2/3}$$

EXAMPLE 16 If $2 \tan^2 \alpha \tan^2 \beta \tan^2 \gamma + \tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma + \tan^2 \gamma \tan^2 \alpha = 1$, prove that $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 1$.

SOLUTION We have,

$$2 \tan^2 \alpha \tan^2 \beta \tan^2 \gamma + \tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma + \tan^2 \gamma \tan^2 \alpha = 1$$

Dividing throughout by $\tan^2 \alpha \tan^2 \beta \tan^2 \gamma$, we get

$$\Rightarrow 2 + \cot^2 \gamma + \cot^2 \alpha + \cot^2 \beta = \cot^2 \alpha \cot^2 \beta \cot^2 \gamma$$

$$\Rightarrow 2 + \operatorname{cosec}^2 \gamma - 1 + \operatorname{cosec}^2 \alpha - 1 + \operatorname{cosec}^2 \beta - 1 = (\operatorname{cosec}^2 \alpha - 1)(\operatorname{cosec}^2 \beta - 1)(\operatorname{cosec}^2 \gamma - 1)$$

$$\Rightarrow \operatorname{cosec}^2 \alpha + \operatorname{cosec}^2 \beta + \operatorname{cosec}^2 \gamma - 1$$

$$= \operatorname{cosec}^2 \alpha \operatorname{cosec}^2 \beta \operatorname{cosec}^2 \gamma - \operatorname{cosec}^2 \alpha \operatorname{cosec}^2 \beta - \operatorname{cosec}^2 \beta \operatorname{cosec}^2 \gamma$$

$$- \operatorname{cosec}^2 \gamma \operatorname{cosec}^2 \alpha + \operatorname{cosec}^2 \alpha + \operatorname{cosec}^2 \beta + \operatorname{cosec}^2 \gamma - 1$$

$$\Rightarrow \operatorname{cosec}^2 \alpha \operatorname{cosec}^2 \beta \operatorname{cosec}^2 \gamma = \operatorname{cosec}^2 \alpha \operatorname{cosec}^2 \beta + \operatorname{cosec}^2 \beta \operatorname{cosec}^2 \gamma + \operatorname{cosec}^2 \gamma \operatorname{cosec}^2 \alpha$$

$$\Rightarrow 1 = \sin^2 \gamma + \sin^2 \alpha + \sin^2 \beta \quad [\text{Multiplying throughout by } \sin^2 \alpha \sin^2 \beta \sin^2 \gamma]$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 1$$

EXAMPLE 17 If $\frac{ax}{\cos \theta} + \frac{by}{\sin \theta} = a^2 - b^2$, and $\frac{ax \sin \theta}{\cos^2 \theta} - \frac{by \cos \theta}{\sin^2 \theta} = 0$, prove that

$$(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}$$

SOLUTION We have,

$$\frac{ax \sin \theta}{\cos^2 \theta} - \frac{by \cos \theta}{\sin^2 \theta} = 0$$

$$\Rightarrow ax \sin^3 \theta - by \cos^3 \theta = 0$$

$$\Rightarrow \frac{\sin^3 \theta}{by} = \frac{\cos^3 \theta}{ax}$$

$$\Rightarrow \left(\frac{\sin^3 \theta}{by} \right)^{2/3} = \left(\frac{\cos^3 \theta}{ax} \right)^{2/3}$$

$$\Rightarrow \frac{\sin^2 \theta}{(by)^{2/3}} = \frac{\cos^2 \theta}{(ax)^{2/3}}$$

$$\Rightarrow \frac{\sin^2 \theta}{(by)^{2/3}} = \frac{\cos^2 \theta}{(ax)^{2/3}} = \frac{\sin^2 \theta + \cos^2 \theta}{(by)^{2/3} + (ax)^{2/3}}$$

[Using ratio and proportions]

$$\Rightarrow \frac{\sin^2 \theta}{(by)^{2/3}} = \frac{\cos^2 \theta}{(ax)^{2/3}} = \frac{1}{(ax)^{2/3} + (by)^{2/3}}$$

$$\Rightarrow \sin^2 \theta = \frac{(by)^{2/3}}{(ax)^{2/3} + (by)^{2/3}} \quad \text{and,} \quad \cos^2 \theta = \frac{(ax)^{2/3}}{(ax)^{2/3} + (by)^{2/3}}$$

$$\Rightarrow \sin \theta = \frac{(by)^{1/3}}{\sqrt{(ax)^{2/3} + (by)^{2/3}}} \quad \text{and,} \quad \cos \theta = \frac{(ax)^{1/3}}{\sqrt{(ax)^{2/3} + (by)^{2/3}}}$$

Substituting these values in $\frac{ax}{\cos \theta} + \frac{by}{\sin \theta} = a^2 - b^2$, we get

$$(ax)^{2/3} \sqrt{(ax)^{2/3} + (by)^{2/3}} + (by)^{2/3} \sqrt{(ax)^{2/3} + (by)^{2/3}} = a^2 - b^2$$

$$\Rightarrow \left\{ \sqrt{(ax)^{2/3} + (by)^{2/3}} \right\} \left\{ (ax)^{2/3} + (by)^{2/3} \right\} = a^2 - b^2$$

$$\Rightarrow \left\{ (ax)^{2/3} + (by)^{2/3} \right\}^{3/2} = a^2 - b^2$$

$$\Rightarrow (ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}$$

EXAMPLE 18 If $m^2 + m'^2 + 2mm' \cos \theta = 1$, $n^2 + n'^2 + 2nn' \cos \theta = 1$, and

$mn + m'n' + (mn' + m'n) \cos \theta = 0$, prove that (i) $m^2 + n^2 = \operatorname{cosec}^2 \theta$ (ii) $m'^2 + n'^2 = \operatorname{cosec}^2 \theta$

SOLUTION (i) We have,

$$m^2 + m'^2 + 2mm' \cos \theta = 1 \quad \text{and} \quad n^2 + n'^2 + 2nn' \cos \theta = 1$$

$$\Rightarrow m'^2 + 2mm' \cos \theta + m^2 \cos^2 \theta - m^2 \cos^2 \theta + m^2 = 1$$

$$\text{and,} \quad n'^2 + 2nn' \cos \theta + n^2 \cos^2 \theta - n^2 \cos^2 \theta + n^2 = 1$$

$$\Rightarrow (m' + m \cos \theta)^2 + m^2 (1 - \cos^2 \theta) = 1 \quad \text{and} \quad (n' + n \cos \theta)^2 + n^2 (1 - \cos^2 \theta) = 1$$

$$\Rightarrow (m' + m \cos \theta)^2 = 1 - m^2 \sin^2 \theta \quad \dots(i) \quad \text{and,} \quad (n' + n \cos \theta)^2 = 1 - n^2 \sin^2 \theta \quad \dots(ii)$$

$$\text{Now,} \quad (m' + m \cos \theta)(n' + n \cos \theta) = m'n' + (mn' + m'n) \cos \theta + mn \cos^2 \theta$$

$$\Rightarrow (m' + m \cos \theta)(n' + n \cos \theta) = -mn + mn \cos^2 \theta \quad [\because mn + m'n' + (mn' + m'n) \cos \theta = 0]$$

$$\Rightarrow (m' + m \cos \theta)(n' + n \cos \theta) = -mn(1 - \cos^2 \theta)$$

$$\Rightarrow (m' + m \cos \theta)(n' + n \cos \theta) = -mn \sin^2 \theta$$

$$\Rightarrow (m' + m \cos \theta)^2 (n' + n \cos \theta)^2 = m^2 n^2 \sin^4 \theta$$

[On squaring both sides]

$$\Rightarrow (1 - m^2 \sin^2 \theta)(1 - n^2 \sin^2 \theta) = m^2 n^2 \sin^4 \theta$$

[Using (i) and (ii)]

$$\Rightarrow 1 - (m^2 + n^2) \sin^2 \theta + m^2 n^2 \sin^4 \theta = m^2 n^2 \sin^4 \theta$$

$$\Rightarrow 1 = (m^2 + n^2) \sin^2 \theta$$

$$\Rightarrow m^2 + n^2 = \operatorname{cosec}^2 \theta$$

(ii) As the given relations do not alter by replacing m by m' and n by n' . Therefore, on replacing m by m' and n by n' in $m^2 + n^2 = \operatorname{cosec}^2 \theta$, we get $m'^2 + n'^2 = \operatorname{cosec}^2 \theta$.

EXAMPLE 19 If $\frac{\sin^4 \theta}{a} + \frac{\cos^4 \theta}{b} = \frac{1}{a+b}$, prove that

$$(i) \frac{\sin^8 \theta}{a^3} + \frac{\cos^8 \theta}{b^3} = \frac{1}{(a+b)^3} \quad (ii) \frac{\sin^{4n} \theta}{a^{2n-1}} + \frac{\cos^{4n} \theta}{b^{2n-1}} = \frac{1}{(a+b)^{2n-1}}, \quad n \in \mathbb{N}$$

SOLUTION We have,

$$\frac{\sin^4 \theta}{a} + \frac{\cos^4 \theta}{b} = \frac{1}{a+b}$$

$$\Rightarrow (a+b) \left(\frac{\sin^4 \theta}{a} + \frac{\cos^4 \theta}{b} \right) = 1$$

$$\Rightarrow (a+b) \left(\frac{\sin^4 \theta}{a} + \frac{\cos^4 \theta}{b} \right) = (\sin^2 \theta + \cos^2 \theta)^2$$

$$\Rightarrow \sin^4 \theta + \cos^4 \theta + \frac{b}{a} \sin^4 \theta + \frac{a}{b} \cos^4 \theta = \sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta$$

$$\Rightarrow \frac{b}{a} \sin^4 \theta + \frac{a}{b} \cos^4 \theta - 2 \sin^2 \theta \cos^2 \theta = 0$$

$$\Rightarrow \left(\sqrt{\frac{b}{a}} \sin^2 \theta - \sqrt{\frac{a}{b}} \cos^2 \theta \right)^2 = 0$$

$$\Rightarrow \sqrt{\frac{b}{a}} \sin^2 \theta = \sqrt{\frac{a}{b}} \cos^2 \theta$$

$$\Rightarrow \tan^2 \theta = \frac{a}{b}$$

$$\Rightarrow \frac{\sin^2 \theta}{a} = \frac{\cos^2 \theta}{b}$$

$$\Rightarrow \frac{\sin^2 \theta}{a} = \frac{\cos^2 \theta}{b} = \frac{\sin^2 \theta + \cos^2 \theta}{a+b}$$

$$\Rightarrow \frac{\sin^2 \theta}{a} = \frac{\cos^2 \theta}{b} = \frac{1}{a+b}$$

$$\Rightarrow \sin^2 \theta = \frac{a}{a+b}, \cos^2 \theta = \frac{b}{a+b} \quad \dots(i)$$

$$(i) \quad \frac{\sin^8 \theta}{a^3} + \frac{\cos^8 \theta}{b^3} = \frac{1}{a^3} (\sin^2 \theta)^4 + \frac{1}{b^3} (\cos^2 \theta)^4$$

$$= \frac{1}{a^3} \left(\frac{a}{a+b} \right)^4 + \frac{1}{b^3} \left(\frac{b}{a+b} \right)^4$$

[Using (i)]

$$= \frac{a}{(a+b)^4} + \frac{b}{(a+b)^4} = \frac{a+b}{(a+b)^4} = \frac{1}{(a+b)^3}$$

$$(ii) \quad \frac{\sin^{4n} \theta}{a^{2n-1}} + \frac{\cos^{4n} \theta}{b^{2n-1}} = \frac{(\sin^2 \theta)^{2n}}{a^{2n-1}} + \frac{(\cos^2 \theta)^{2n}}{b^{2n-1}}$$

$$= \frac{1}{a^{2n-1}} \left(\frac{a}{a+b} \right)^{2n} + \frac{1}{b^{2n-1}} \left(\frac{b}{a+b} \right)^{2n}$$

$$= \frac{a}{(a+b)^{2n}} + \frac{b}{(a+b)^{2n}} = \frac{a+b}{(a+b)^{2n}} = \frac{1}{(a+b)^{2n-1}}$$

EXAMPLE 20 If $\frac{\cos^4 \alpha}{\cos^2 \beta} + \frac{\sin^4 \alpha}{\sin^2 \beta} = 1$, prove that

$$(i) \sin^4 \alpha + \sin^4 \beta = 2 \sin^2 \alpha \sin^2 \beta$$

$$(ii) \frac{\cos^4 \beta}{\cos^2 \alpha} + \frac{\sin^4 \beta}{\sin^2 \alpha} = 1$$

SOLUTION We have,

$$\frac{\cos^4 \alpha}{\cos^2 \beta} + \frac{\sin^4 \alpha}{\sin^2 \beta} = 1$$

$$\Rightarrow \cos^4 \alpha \sin^2 \beta + \sin^4 \alpha \cos^2 \beta = \cos^2 \beta \sin^2 \beta$$

$$\Rightarrow \cos^4 \alpha (1 - \cos^2 \beta) + \cos^2 \beta (1 - \cos^2 \alpha)^2 = \cos^2 \beta (1 - \cos^2 \beta)$$

$$\Rightarrow \cos^4 \alpha - \cos^4 \alpha \cos^2 \beta + \cos^2 \beta - 2 \cos^2 \alpha \cos^2 \beta + \cos^4 \alpha \cos^2 \beta = \cos^2 \beta - \cos^4 \beta$$

$$\Rightarrow \cos^4 \alpha - 2 \cos^2 \alpha \cos^2 \beta + \cos^4 \beta = 0$$

$$\Rightarrow (\cos^2 \alpha - \cos^2 \beta)^2 = 0$$

$$\Rightarrow \cos^2 \alpha - \cos^2 \beta = 0$$

$$\Rightarrow \cos^2 \alpha = \cos^2 \beta$$

...(i)

$$\Rightarrow 1 - \sin^2 \alpha = 1 - \sin^2 \beta$$

$$\Rightarrow \sin^2 \alpha = \sin^2 \beta$$

... (ii)

$$(i) \quad \sin^4 \alpha + \sin^4 \beta = (\sin^2 \alpha - \sin^2 \beta)^2 + 2 \sin^2 \alpha \sin^2 \beta$$

$$= 2 \sin^2 \alpha \sin^2 \beta$$

$$[\because \sin^2 \alpha = \sin^2 \beta]$$

$$(ii) \quad \frac{\cos^4 \beta}{\cos^2 \alpha} + \frac{\sin^4 \beta}{\sin^2 \alpha} = \frac{\cos^2 \beta \cos^2 \beta}{\cos^2 \alpha} + \frac{\sin^2 \beta \sin^2 \beta}{\sin^2 \alpha}$$

$$= \frac{\cos^2 \beta \cos^2 \alpha}{\cos^2 \alpha} + \frac{\sin^2 \beta \sin^2 \alpha}{\sin^2 \alpha}$$

[Using (i) and (ii)]

$$= \cos^2 \beta + \sin^2 \beta$$

EXAMPLE 21 If x is any non-zero real number, show that $\cos \theta$ and $\sin \theta$ can never be equal to $x + \frac{1}{x}$.

SOLUTION We have following cases:

CASE I When $x > 0$: In this case, we have

$$x + \frac{1}{x} = (\sqrt{x})^2 + \left(\frac{1}{\sqrt{x}}\right)^2 - 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}} + 2 \times \sqrt{x} \times \frac{1}{\sqrt{x}}$$

$$\Rightarrow x + \frac{1}{x} = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 + 2 \geq 2$$

$$\left[\because \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0 \right]$$

CASE II When $x < 0$: Let $x = -y$. Then, $y > 0$

$$\therefore x + \frac{1}{x} = -y - \frac{1}{y} = -\left(y + \frac{1}{y}\right)$$

$$\text{But, } y + \frac{1}{y} \geq 2$$

[From Case I]

$$\Rightarrow -\left(y + \frac{1}{y}\right) \leq -2 \Rightarrow x + \frac{1}{x} \leq -2$$

$$\therefore x + \frac{1}{x} \geq 2 \quad \text{for } x > 0 \text{ and, } x + \frac{1}{x} \leq -2 \quad \text{for } x < 0$$

But, $-1 \leq \sin \theta \leq 1$ and $-1 \leq \cos \theta \leq 1$ for all θ .

Hence, $\sin \theta$ and $\cos \theta$ cannot be equal to $x + \frac{1}{x}$ for any non-zero x .

EXAMPLE 22 If $A = \cos^2 \theta + \sin^4 \theta$, prove that $\frac{3}{4} \leq A \leq 1$ for all values of θ .

SOLUTION We have

$$A = \cos^2 \theta + \sin^4 \theta = \cos^2 \theta + (\sin^2 \theta)^2$$

Now, $-1 \leq \sin \theta \leq 1$ for all θ

$$\Rightarrow 0 \leq \sin^2 \theta \leq 1 \text{ for all } \theta$$

$$\Rightarrow (\sin^2 \theta)^2 \leq \sin^2 \theta$$

[For $0 < x < 1$, $x^n < x$ for all $n \in \mathbb{N} - \{1\}$]

$$\Rightarrow \cos^2 \theta + (\sin^2 \theta)^2 \leq \cos^2 \theta + \sin^2 \theta \text{ for all } \theta$$

$$\Rightarrow A \leq 1 \text{ for all } \theta$$

...(i)

Again,

$$A = \cos^2 \theta + \sin^4 \theta$$

$$\Rightarrow A = 1 - \sin^2 \theta + (\sin^2 \theta)^2$$

$$\Rightarrow A = 1 - \frac{1}{4} + \left\{ \frac{1}{4} - \sin^2 \theta + (\sin^2 \theta)^2 \right\}$$

$$\Rightarrow A = \frac{3}{4} + \left(\frac{1}{2} - \sin^2 \theta \right)^2$$

Now,

$$\left(\frac{1}{2} - \sin^2 \theta \right)^2 \geq 0 \text{ for all } \theta$$

$$\Rightarrow \frac{3}{4} + \left(\frac{1}{2} - \sin^2 \theta \right)^2 \geq \frac{3}{4} \text{ for all } \theta$$

$$\Rightarrow A \geq \frac{3}{4} \text{ for all } \theta$$

...(ii)

From (i) and (ii), we obtain $\frac{3}{4} \leq A \leq 1$ for all θ

EXERCISE 5.1

LEVEL-1

Prove the following identities (1-16)

$$1. \sec^4 \theta - \sec^2 \theta = \tan^4 \theta + \tan^2 \theta$$

$$2. \sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$$

$$3. (\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta) = 1$$

$$4. \operatorname{cosec} \theta (\sec \theta - 1) - \cot \theta (1 - \cos \theta) = \tan \theta - \sin \theta$$

$$5. \frac{1 - \sin A \cos A}{\cos A (\sec A - \operatorname{cosec} A)} \cdot \frac{\sin^2 A - \cos^2 A}{\sin^3 A + \cos^3 A} = \sin A$$

$$6. \frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} = (\sec A \operatorname{cosec} A + 1)$$

$$7. \frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$$

$$8. (\sec A \sec B + \tan A \tan B)^2 - (\sec A \tan B + \tan A \sec B)^2 = 1$$

$$9. \frac{\cos \theta}{1 - \sin \theta} = \frac{1 + \cos \theta + \sin \theta}{1 + \cos \theta - \sin \theta}$$

$$10. \frac{\tan^3 \theta}{1 + \tan^2 \theta} + \frac{\cot^3 \theta}{1 + \cot^2 \theta} = \frac{1 - 2 \sin^2 \theta \cos^2 \theta}{\sin \theta \cos \theta}$$

$$11. 1 - \frac{\sin^2 \theta}{1 + \cot \theta} + \frac{\cos^2 \theta}{1 + \tan \theta} = \sin \theta \cos \theta$$

$$12. \left(\frac{1}{\sec^2 \theta - \cos^2 \theta} + \frac{1}{\operatorname{cosec}^2 \theta - \sin^2 \theta} \right) \sin^2 \theta \cos^2 \theta = \frac{1 - \sin^2 \theta \cos^2 \theta}{2 + \sin^2 \theta \cos^2 \theta}$$

$$13. (1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2 = \sec^2 \alpha \sec^2 \beta$$

$$14. \frac{(1 + \cot \theta + \tan \theta)(\sin \theta - \cos \theta)}{\sec^3 \theta - \operatorname{cosec}^3 \theta} = \sin^2 \theta \cos^2 \theta$$

$$15. \frac{2 \sin \theta \cos \theta - \cos \theta}{1 - \sin \theta + \sin^2 \theta - \cos^2 \theta} = \cot \theta$$

$$16. \cos \theta (\tan \theta + 2)(2 \tan \theta + 1) = 2 \sec \theta + 5 \sin \theta$$

$$17. \text{ If } x = \frac{2 \sin \theta}{1 + \cos \theta + \sin \theta}, \text{ then prove that } \frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} \text{ is also equal to } x.$$

[NCERT EXEMPLAR]

$$18. \text{ If } \sin \theta = \frac{a^2 - b^2}{a^2 + b^2}, \text{ find the values of } \tan \theta, \sec \theta \text{ and } \operatorname{cosec} \theta$$

$$19. \text{ If } \tan x = \frac{b}{a}, \text{ then find the value of } \sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}}.$$

[NCERT EXEMPLAR]

$$20. \text{ If } \tan \theta = \frac{a}{b}, \text{ show that } \frac{a \sin \theta - b \cos \theta}{a \sin \theta + b \cos \theta} = \frac{a^2 - b^2}{a^2 + b^2}$$

LEVEL-2

$$21. \text{ If } \operatorname{cosec} \theta - \sin \theta = a^3, \sec \theta - \cos \theta = b^3, \text{ then prove that } a^2 b^2 (a^2 + b^2) = 1.$$

$$22. \text{ If } \cot \theta (1 + \sin \theta) = 4m \text{ and } \cot \theta (1 - \sin \theta) = 4n, \text{ prove that } (m^2 - n^2)^2 = mn.$$

$$23. \text{ If } \sin \theta + \cos \theta = m, \text{ then prove that}$$

$$\sin^6 \theta + \cos^6 \theta = \frac{4 - 3(m^2 - 1)^2}{4}, \text{ where } m^2 \leq 2$$

$$24. \text{ If } a = \sec \theta - \tan \theta \text{ and } b = \operatorname{cosec} \theta + \cot \theta, \text{ then show that } ab + a - b + 1 = 0.$$

[NCERT EXEMPLAR]

$$25. \text{ Prove that: } \left| \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} \right| = -\frac{2}{\cos \theta}, \text{ where } \frac{\pi}{2} < \theta < \pi$$

$$26. \text{ If } T_n = \sin^n \theta + \cos^n \theta, \text{ prove that}$$

$$(i) \frac{T_3 - T_5}{T_1} = \frac{T_5 - T_7}{T_3} \quad (ii) 2T_6 - 3T_4 + 1 = 0 \quad (iii) 6T_{10} - 15T_8 + 10T_6 - 1 = 0$$

ANSWERS

$$18. \tan \theta = \frac{a^2 - b^2}{2ab}, \sec \theta = \frac{a^2 + b^2}{2ab}, \operatorname{cosec} \theta = \frac{a^2 + b^2}{a^2 - b^2}$$

$$19. \frac{2 \cos x}{\sqrt{\cos^2 x - \sin^2 x}}$$

HINTS TO SELECTED PROBLEMS

21. We have, $\operatorname{cosec} \theta - \sin \theta = a^3$, $\sec \theta - \cos \theta = b^3$

$$\Rightarrow \frac{1 - \sin^2 \theta}{\sin \theta} = a^3, \frac{1 - \cos^2 \theta}{\cos \theta} = b^3$$

$$\Rightarrow \frac{\cos^2 \theta}{\sin \theta} = a^3, \frac{\sin^2 \theta}{\cos \theta} = b^3$$

$$\Rightarrow \frac{\sin^2 \theta}{\cos \theta} \div \frac{\cos^2 \theta}{\sin \theta} = \frac{b^3}{a^3}$$

$$\Rightarrow \tan^3 \theta = \frac{b^3}{a^3} \Rightarrow \tan \theta = \frac{b}{a} \Rightarrow \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \text{ and } \cos \theta = \frac{a}{\sqrt{a^2 + b^2}}$$

Substituting these values of $\sin \theta$ and $\cos \theta$ in $\frac{\cos^2 \theta}{\sin \theta} = a^3$, we obtain

$$\frac{a^2}{b \sqrt{(a^2 + b^2)}} = a^3 \Rightarrow ab \sqrt{a^2 + b^2} = 1 \Rightarrow a^2 b^2 (a^2 + b^2) = 1$$

22. We have, $\cot \theta (1 + \sin \theta) = 4m$ and $\cot \theta (1 - \sin \theta) = 4n$

$$\Rightarrow \cot \theta + \cos \theta = 4m \text{ and } \cot \theta - \cos \theta = 4n$$

$$\Rightarrow (\cot \theta + \cos \theta)^2 - (\cot \theta - \cos \theta)^2 = 16m^2 - 16n^2 \text{ and } (\cot \theta + \cos \theta)(\cot \theta - \cos \theta) = 16mn$$

$$\Rightarrow 4 \cot \theta \cos \theta = 16(m^2 - n^2) \text{ and } \cot^2 \theta - \cos^2 \theta = 16mn$$

$$\Rightarrow \frac{\cos^2 \theta}{\sin \theta} = 4(m^2 - n^2) \text{ and } \frac{\cos^4 \theta}{\sin^2 \theta} = 16mn$$

$$\Rightarrow \frac{\cos^4 \theta}{\sin^2 \theta} = 16(m^2 - n^2)^2 \text{ and } \frac{\cos^4 \theta}{\sin^2 \theta} = 16mn$$

$$\Rightarrow 16(m^2 - n^2)^2 = 16mn$$

$$\Rightarrow (m^2 - n^2) = mn$$

5.4 SIGNS OF THE TRIGONOMETRIC RATIOS OR FUNCTIONS

In the previous section, we have introduced six trigonometric ratios. Their signs depend on the quadrant in which the terminal side of the angle lies. We always take the length $OP = r$ (see Fig. 5.1 — 5.4) to be positive.

Thus, $\sin \theta = \frac{y}{r}$ has the sign of y , $\cos \theta = \frac{x}{r}$ has the sign of x . The sign of $\tan \theta$ depends on the signs of x and y and similarly the signs of other trigonometric ratios are decided by the signs of x and/or y . Thus, we have the following :

In first quadrant: We have, $x > 0$ and $y > 0$

$$\therefore \sin \theta = \frac{y}{r} > 0, \cos \theta = \frac{x}{r} > 0, \tan \theta = \frac{y}{x} > 0, \operatorname{cosec} \theta = \frac{r}{y} > 0, \sec \theta = \frac{r}{x} > 0 \text{ and } \cot \theta = \frac{x}{y} > 0.$$

Thus, in the first quadrant all trigonometric functions are positive.

In second quadrant: We have, $x < 0$ and $y > 0$

$$\therefore \sin \theta = \frac{y}{r} > 0, \cos \theta = \frac{x}{r} < 0, \tan \theta = \frac{y}{x} < 0, \operatorname{cosec} \theta = \frac{r}{y} > 0, \sec \theta = \frac{r}{x} < 0 \text{ and } \cot \theta = \frac{x}{y} < 0.$$

Thus, in the second quadrant sine and cosecant functions are positive and all others are negative.

In third quadrant: We have, $x < 0$ and $y < 0$

$$\therefore \sin \theta = \frac{y}{r} < 0, \cos \theta = \frac{x}{r} < 0, \tan \theta = \frac{y}{x} > 0, \operatorname{cosec} \theta = \frac{r}{y} < 0, \sec \theta = \frac{r}{x} < 0 \text{ and } \cot \theta = \frac{x}{y} > 0.$$

Thus, in the third quadrant all trigonometric functions are negative except tangent and its reciprocal cotangent.

In fourth quadrant: We have, $x > 0$ and $y < 0$

$$\therefore \sin \theta = \frac{y}{r} < 0, \cos \theta = \frac{x}{r} > 0, \tan \theta = \frac{y}{x} < 0, \operatorname{cosec} \theta = \frac{r}{y} < 0, \sec \theta = \frac{r}{x} > 0 \text{ and } \cot \theta = \frac{x}{y} < 0.$$

Thus, in the fourth quadrant all trigonometric functions are negative except cosine and its reciprocal secant.

It follows from the above discussion that the signs of the trigonometric ratios in different quadrants are as under :

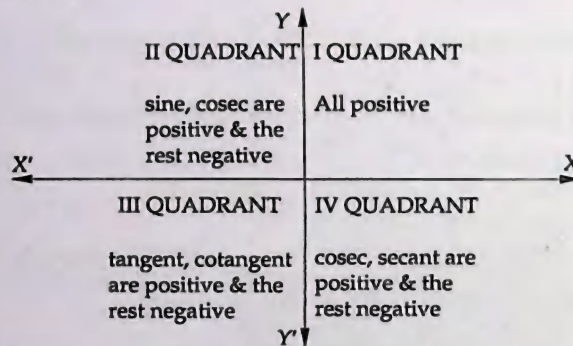


Fig. 5.6

SIMPLE RULE TO REMEMBER A crude aid to memorise the signs of trigonometrical ratios in different quadrants is the four-word phrase "ALL SCHOOL TO COLLEGE". The first letter of the first word in this phrase is 'A'. This may be taken to indicate that all trigonometric ratios are positive in the first quadrant. The first letter of the second word is 'S'. This indicates that sine and its reciprocal are positive in the second quadrant. The first letter of third word is 'T'. This may be taken as to indicate that tangent and its reciprocal are positive in the third quadrant. The first letter of the fourth word in the phrase is 'C' which may be taken as to indicate that only cosine and its reciprocal are positive in the fourth quadrant.

5.5 VARIATIONS IN VALUES OF TRIGONOMETRIC FUNCTIONS IN DIFFERENT QUADRANTS

DIFFERENT QUADRANTS Let $X'OX$ and YOY' be the coordinate axes. Draw a circle with centre at origin O and radius unity. Suppose the circle cuts the coordinate axes at A, B, A' and B' as shown in Fig. 5.7. Let $P(x, y)$ be a point on the circle such that $\angle AOP = \theta$. Then, $x = \cos \theta$ and $y = \sin \theta$.

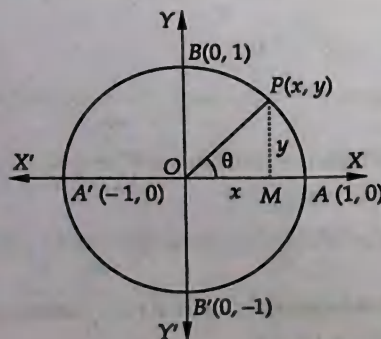


Fig. 5.7

It is evident from the Fig. 5.7 that

$$-1 \leq x \leq 1 \text{ and } -1 \leq y \leq 1$$

$\Rightarrow -1 \leq \cos \theta \leq 1$ and $-1 \leq \sin \theta \leq 1$ for all values of θ .

In the first quadrant as the angle θ increases from 0° to 90° , we observe that the values of y increase from 0 to 1. Consequently $\sin \theta$ increases from 0 to 1. In the second quadrant as θ increases from 90° to 180° , y decreases from 1 to 0. Consequently, $\sin \theta$ decreases from 1 to 0. In the third quadrant as θ increases from 180° to 270° , $\sin \theta (= y)$ decreases from 0 to -1 and finally in the fourth quadrant $\sin \theta$ increases from -1 to 0 as θ increases from 270° to 360° . Similarly, we can observe the variations in the values of other trigonometric functions. The following table exhibits the variations in the values of all trigonometrical ratios.

I QUADRANT		II QUADRANT	
$\sin \theta$	increases from 0 to 1	$\sin \theta$	decreases from 1 to 0
$\cos \theta$	decreases from 1 to 0	$\cos \theta$	decreases from 0 to -1
$\tan \theta$	increases from 0 to ∞	$\tan \theta$	increases from $-\infty$ to 0
$\cot \theta$	decreases from ∞ to 0	$\cot \theta$	decreases from 0 to $-\infty$
$\sec \theta$	increases from 1 to ∞	$\sec \theta$	increases from $-\infty$ to -1
$\operatorname{cosec} \theta$	decreases from ∞ to 1	$\operatorname{cosec} \theta$	decreases from 1 to ∞
III QUADRANT		IV QUADRANT	
$\sin \theta$	decreases from 0 to -1	$\sin \theta$	increases from -1 to 0
$\cos \theta$	decreases from -1 to 0	$\cos \theta$	increases from 0 to 1
$\tan \theta$	increases from 0 to ∞	$\tan \theta$	increases from $-\infty$ to 0
$\cot \theta$	decreases from ∞ to 0	$\cot \theta$	decreases from 0 to $-\infty$
$\sec \theta$	decreases from -1 to $-\infty$	$\sec \theta$	decreases from ∞ to 1
$\operatorname{cosec} \theta$	decreases from $-\infty$ to -1	$\operatorname{cosec} \theta$	decreases from -1 to ∞

REMARK Note that $+\infty$ and $-\infty$ are two symbols. These are not real numbers. When we say that $\tan \theta$ increases from 0 to ∞ as θ varies from 0 to $\pi/2$, it means that $\tan \theta$ increases in the interval $(0, \pi/2)$ and it attains arbitrarily large positive values as θ tends to $\pi/2$. Similarly, we interpret for other trigonometrical functions.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find $\sin \theta$ and $\tan \theta$ if $\cos \theta = -\frac{12}{13}$ and θ lies in the third quadrant.

SOLUTION We have,

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \sin \theta = \pm \sqrt{1 - \cos^2 \theta}$$

In third quadrant $\sin \theta$ is negative.

$$\therefore \sin \theta = -\sqrt{1 - \cos^2 \theta} \Rightarrow \sin \theta = -\sqrt{1 - \left(-\frac{12}{13}\right)^2} = -\frac{5}{13}$$

$$\text{Now, } \tan \theta = \frac{\sin \theta}{\cos \theta} \Rightarrow \tan \theta = -\frac{5}{13} \times \frac{13}{-12} = \frac{5}{12}$$

EXAMPLE 2 Find the values of $\cos \theta$ and $\tan \theta$, if $\sin \theta = -\frac{3}{5}$ and $\pi < \theta < \frac{3\pi}{2}$.

SOLUTION We have,

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

In the third quadrant $\cos \theta$ is negative and $\tan \theta$ is positive.

$$\therefore \cos \theta = -\sqrt{1 - \sin^2 \theta} \Rightarrow \cos \theta = -\sqrt{1 - \frac{9}{25}} = -\frac{4}{5}$$

$$\text{and, } \tan \theta = \frac{\sin \theta}{\cos \theta} \Rightarrow \tan \theta = -\frac{3}{5} \times -\frac{5}{4} = \frac{3}{4}$$

EXAMPLE 3 Find all other trigonometrical ratios if $\sin \theta = -\frac{2\sqrt{6}}{5}$ and θ lies in quadrant III.

SOLUTION We have,

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

In the third quadrant $\cos \theta$ is negative.

$$\therefore \cos \theta = -\sqrt{1 - \sin^2 \theta} \Rightarrow \cos \theta = -\sqrt{1 - \frac{24}{25}} = -\frac{1}{5}$$

In the third quadrant $\tan \theta$ is positive.

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta} \Rightarrow \tan \theta = -\frac{2\sqrt{6}}{5} \times -\frac{5}{1} = 2\sqrt{6}$$

$$\text{Now, } \operatorname{cosec} \theta = \frac{1}{\sin \theta} \Rightarrow \operatorname{cosec} \theta = -\frac{5}{2\sqrt{6}}$$

$$\sec \theta = \frac{1}{\cos \theta} \Rightarrow \sec \theta = -5$$

$$\text{and, } \cot \theta = \frac{1}{\tan \theta} \Rightarrow \cot \theta = \frac{1}{2\sqrt{6}}$$

EXAMPLE 4 If $\cos \theta = -\frac{1}{2}$ and $\pi < \theta < \frac{3\pi}{2}$, find the value of $4 \tan^2 \theta - 3 \operatorname{cosec}^2 \theta$.

SOLUTION Since θ lies in the third quadrant. Therefore, $\sin \theta$ is negative and $\tan \theta$ is positive.

$$\text{Now, } \sin \theta = \pm \sqrt{1 - \cos^2 \theta} \Rightarrow \sin \theta = -\sqrt{1 - \frac{1}{4}} = -\frac{\sqrt{3}}{2} \Rightarrow \operatorname{cosec} \theta = \frac{-2}{\sqrt{3}}$$

$$\text{And, } \tan \theta = \frac{\sin \theta}{\cos \theta} = \sqrt{3}.$$

$$\text{Hence, } 4 \tan^2 \theta - 3 \operatorname{cosec}^2 \theta = 4 \times 3 - 3 \times \frac{4}{3} = 8.$$

EXAMPLE 5 If $\sec \theta = \sqrt{2}$ and $\frac{3\pi}{2} < \theta < 2\pi$, find the value of $\frac{1 + \tan \theta + \operatorname{cosec} \theta}{1 + \cot \theta - \operatorname{cosec} \theta}$.

SOLUTION We have,

$$\sec \theta = \sqrt{2} \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

$$\therefore \sin \theta = \pm \sqrt{1 - \cos^2 \theta} = \pm \sqrt{1 - \frac{1}{2}} = \pm \frac{1}{\sqrt{2}}$$

But, θ lies in the fourth quadrant in which $\sin \theta$ is negative.

$$\therefore \sin \theta = -\frac{1}{\sqrt{2}} \Rightarrow \operatorname{cosec} \theta = -\sqrt{2}$$

$$\text{Now, } \tan \theta = \frac{\sin \theta}{\cos \theta} \Rightarrow \tan \theta = -\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{1} = -1 \Rightarrow \cot \theta = -1$$

$$\therefore \frac{1 + \tan \theta + \operatorname{cosec} \theta}{1 + \cot \theta - \operatorname{cosec} \theta} = \frac{1 - 1 - \sqrt{2}}{1 - 1 + \sqrt{2}} = -1$$

LEVEL-2

EXAMPLE 6 ✓ Prove that :

$$\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \begin{cases} \sec \theta - \tan \theta, & \text{if } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ -\sec \theta + \tan \theta, & \text{if } \frac{\pi}{2} < \theta < \frac{3\pi}{2} \end{cases}$$

SOLUTION We have,

$$\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \sqrt{\frac{(1 - \sin \theta)^2}{1 - \sin^2 \theta}} = \frac{1 - \sin \theta}{\sqrt{\cos^2 \theta}} = \frac{1 - \sin \theta}{|\cos \theta|}$$

$$\Rightarrow \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \begin{cases} \frac{1 - \sin \theta}{\cos \theta}, & \text{if } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ \frac{1 - \sin \theta}{-\cos \theta}, & \text{if } \frac{\pi}{2} < \theta < \frac{3\pi}{2} \end{cases}$$

$$\Rightarrow \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \begin{cases} \sec \theta - \tan \theta, & \text{if } -\frac{\pi}{2} < \theta < \frac{\pi}{2} \\ -\sec \theta + \tan \theta, & \text{if } \frac{\pi}{2} < \theta < \frac{3\pi}{2} \end{cases}$$

EXAMPLE 7 ✓ Prove that : $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \begin{cases} \operatorname{cosec} \theta + \cot \theta, & \text{if } 0 < \theta < \pi \\ -\operatorname{cosec} \theta - \cot \theta, & \text{if } \pi < \theta < 2\pi \end{cases}$

SOLUTION We have,

$$\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \sqrt{\frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta}}$$

$$\Rightarrow \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \frac{1 + \cos \theta}{\sqrt{\sin^2 \theta}} = \frac{1 + \cos \theta}{|\sin \theta|}$$

$$\Rightarrow \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \begin{cases} \frac{1 + \cos \theta}{\sin \theta}, & \text{if } 0 < \theta < \pi \\ \frac{1 + \cos \theta}{-\sin \theta}, & \text{if } \pi < \theta < 2\pi \end{cases}$$

$$\Rightarrow \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \begin{cases} \operatorname{cosec} \theta + \cot \theta, & \text{if } 0 < \theta < \pi \\ -\operatorname{cosec} \theta - \cot \theta, & \text{if } \pi < \theta < 2\pi \end{cases}$$

EXAMPLE 8 ✓ Prove that :

$$\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = \begin{cases} \frac{2}{\cos \theta}, & \text{if } 0 \leq \theta < \frac{\pi}{2} \\ -\frac{2}{\cos \theta}, & \text{if } \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

SOLUTION We have,

$$\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = \frac{(1 - \sin \theta) + (1 + \sin \theta)}{\sqrt{1 - \sin^2 \theta}}$$

$$\Rightarrow \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} + \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = \frac{2}{\sqrt{\cos^2 \theta}} = \frac{2}{|\cos \theta|} \quad [\because \sqrt{x^2} = |x|]$$

$$\Rightarrow \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} + \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = \begin{cases} \frac{2}{\cos \theta}, & \text{if } 0 \leq \theta < \frac{\pi}{2} \\ -\frac{2}{\cos \theta}, & \text{if } \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

EXERCISE 5.2**LEVEL-1** ✓

1. Find the values of the other five trigonometric functions in each of the following:

(i) $\cot \theta = \frac{12}{5}$, θ in quadrant III

(ii) $\cos \theta = -\frac{1}{2}$, θ in quadrant II

(iii) $\tan \theta = \frac{3}{4}$, θ in quadrant III

(iv) $\sin \theta = \frac{3}{5}$, θ in quadrant I

2. If $\sin \theta = \frac{12}{13}$ and θ lies in the second quadrant, find the value of $\sec \theta + \tan \theta$.

3. If $\sin \theta = \frac{3}{5}$, $\tan \phi = \frac{1}{2}$ and $\frac{\pi}{2} < \theta < \pi < \phi < \frac{3\pi}{2}$, find the value of $8 \tan \theta - \sqrt{5} \sec \phi$.

4. If $\sin \theta + \cos \theta = 0$ and θ lies in the fourth quadrant, find $\sin \theta$ and $\cos \theta$.

5. If $\cos \theta = -\frac{3}{5}$ and $\pi < \theta < \frac{3\pi}{2}$, find the values of other five trigonometric functions and hence evaluate $\frac{\operatorname{cosec} \theta + \cot \theta}{\sec \theta - \tan \theta}$.

ANSWERS

1. (i) $\sin \theta = -\frac{5}{13}$, $\cos \theta = -\frac{12}{13}$, $\tan \theta = \frac{5}{12}$, $\operatorname{cosec} \theta = -\frac{13}{5}$, $\sec \theta = -\frac{13}{12}$

(ii) $\sin \theta = \frac{\sqrt{3}}{2}$, $\tan \theta = -\sqrt{3}$, $\operatorname{cosec} \theta = \frac{2}{\sqrt{3}}$, $\cot \theta = \frac{-1}{\sqrt{3}}$, $\sec \theta = -2$

(iii) $\sin \theta = -\frac{3}{5}$, $\cos \theta = -\frac{4}{5}$, $\operatorname{cosec} \theta = -\frac{5}{3}$, $\sec \theta = -\frac{5}{4}$, $\cot \theta = \frac{4}{3}$

(iv) $\cos \theta = \frac{4}{5}$, $\tan \theta = \frac{3}{4}$, $\sec \theta = \frac{5}{4}$, $\cot \theta = \frac{4}{3}$, $\operatorname{cosec} \theta = \frac{5}{3}$

2. -5 3. $-\frac{7}{2}$ 4. $-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ 5. $\frac{1}{6}$

HINTS TO SELECTED PROBLEM

4. We have,

$$\sin \theta + \cos \theta = 0 \Rightarrow \sin \theta = -\cos \theta \Rightarrow \tan \theta = -1$$

$$\therefore \sec^2 \theta = 1 + \tan^2 \theta \Rightarrow \sec^2 \theta = 1 + (-1)^2 = 2 \Rightarrow \sec \theta = \sqrt{2} \Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

5.6 TRIGONOMETRIC RATIOS OF ALLIED ANGLES

Two angles are said to be allied when their sum or difference is either zero or a multiple of 90° .

The angles $-\theta$, $90^\circ \pm \theta$, $180^\circ \pm \theta$, $360^\circ \pm \theta$ etc are angles allied to the angle θ if θ is measured in degrees. However, if θ is measured in radians, then the angles allied to θ are $-\theta$, $\frac{\pi}{2} \pm \theta$, $\pi \pm \theta$, $2\pi \pm \theta$ etc. Using trigonometric ratios of allied angles we can find the trigonometric ratios of angles of any magnitude.

5.6.1 TRIGONOMETRIC RATIOS OF $(-\theta)$ IN TERMS OF THAT OF θ

Let a revolving ray starting from its initial position OX , trace out an angle $\angle XO A = \theta$. Let $P(x, y)$ be a point on OA such that $OP = r$. Draw PM perpendicular from P on x -axis. Let there be another revolving ray OA' which starts from the initial position OX and describes an angle $\angle XO A' = -\theta$ in the clockwise sense. Let P' be a point on OA' such that $OP' = OP$. Draw $P'M'$ perpendicular from P' on x -axis. Clearly, M and M' coincide and $\triangle OMP$ is congruent to $\triangle OMP'$. Then the coordinates of P' are $(x, -y)$.

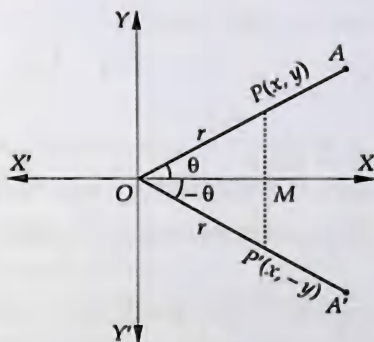


Fig. 5.8

$$\therefore \sin(-\theta) = \frac{-y}{r} = -\frac{y}{r} = -\sin \theta, \cos(-\theta) = \frac{x}{r} = \cos \theta, \tan(-\theta) = \frac{-y}{x} = -\tan \theta.$$

Taking the reciprocals of these trigonometric ratios, we have

$$\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta, \sec(-\theta) = \sec \theta \text{ and } \cot(-\theta) = -\cot \theta.$$

5.6.2 TRIGONOMETRIC RATIOS OF $(90^\circ - \theta)$ IN TERMS OF THAT OF θ

Let a revolving ray starting from its initial position OX , trace out an angle $\angle XO A = \theta$. Let $P(x, y)$ be a point on OA such that $OP = r$. Draw PM perpendicular from P on x -axis. Let another revolving ray OA' starting from the initial position OX , trace out angle of 90° to coincide with OY and then it rotates in the clockwise direction through an angle θ . Then, OA' in its final position traces out an angle $\angle XO A' = 90^\circ - \theta$. Let P' be a point on OA' such that $OP' = OP = r$. From P' draw perpendicular $P'M'$ on x -axis. Then, $\triangle OMP$ and $\triangle OM'P'$ are congruent.

Clearly, $M'P' = OM = x$ and $OM' = MP = y$

So, the coordinates of P' are (y, x) .

$$\therefore \sin(90^\circ - \theta) = \frac{P'M'}{OP'} = \frac{x}{r} = \cos \theta.$$

$$\cos(90^\circ - \theta) = \frac{OM'}{OP'} = \frac{y}{r} = \sin \theta$$

$$\tan(90^\circ - \theta) = \frac{P'M'}{OM'} = \frac{x}{y} = \cot \theta$$

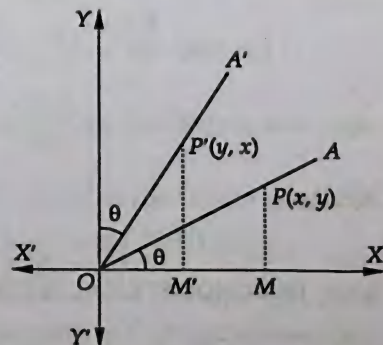


Fig. 5.9

Similarly, we obtain that

$$\operatorname{cosec}(90^\circ - \theta) = \sec \theta, \sec(90^\circ - \theta) = \operatorname{cosec} \theta \text{ and } \cot(90^\circ - \theta) = \tan \theta.$$

5.6.3 TRIGONOMETRIC RATIOS OF $(90^\circ + \theta)$ IN TERMS OF THAT OF θ

Let a revolving ray OA starting from its initial position OX , trace out an angle $\angle XO A = \theta$ and let another revolving ray OA' starting from the same initial position OX , first trace out an angle θ so as to coincide with OA and then it revolves through an angle of 90° in anticlockwise direction to form an angle $\angle XO A' = 90^\circ + \theta$. Let P and P' be points on OA and OA' respectively such that $OP = OP' = r$. Draw perpendiculars PM and $P'M'$ from P and P' respectively on OX . Let the coordinates of P be (x, y) . Then, $OM = x$ and $P'M' = y$.

Clearly, $\triangle P'M'O$ is congruent to $\triangle OMP$.

$\therefore OM' = PM = y$ and $P'M' = OM = x$.

So, the coordinates of P' are $(-y, x)$.

Hence,

$$\sin(90^\circ + \theta) = \frac{M'P'}{OP'} = \frac{x}{r} = \cos \theta.$$

$$\cos(90^\circ + \theta) = \frac{OM'}{OP'} = \frac{-y}{r} = -\sin \theta$$

$$\tan(90^\circ + \theta) = \frac{M'P'}{OM'} = \frac{x}{-y} = -\cot \theta$$

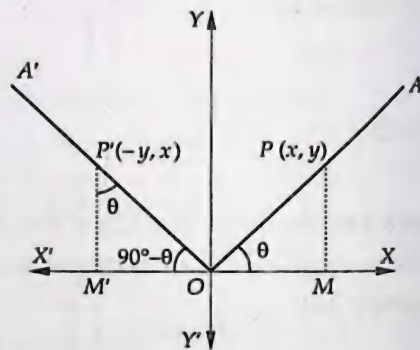


Fig. 5.10

Similarly, we obtain

$$\operatorname{cosec}(90^\circ + \theta) = \sec \theta, \sec(90^\circ + \theta) = -\operatorname{cosec} \theta \text{ and } \cot(90^\circ + \theta) = -\tan \theta.$$

5.6.4 TRIGONOMETRIC RATIOS OF $(180^\circ - \theta)$ IN TERMS OF THAT OF θ

Let a revolving ray OA starting from its initial position OX , trace out an angle $\angle XO A = \theta$. Let another revolving ray OA' starting from its initial position OX , trace out angle of 180° to coincide with OX' and then rotates in clockwise sense through an angle θ . Let P and P' be points on OA and OA' respectively such that $OP = OP' = r$. Draw perpendiculars PM and $P'M'$ from P and P' on OX and OX' respectively.

Clearly, triangles OMP and $OM'P'$ are congruent.

$\therefore M'P' = MP = y$ and $OM' = OM$.

Hence, the coordinates of P' are $(-x, y)$.

$$\text{Now, } \sin(180^\circ - \theta) = \frac{P'M'}{OP'} = \frac{y}{r} = \sin \theta.$$

$$\cos(180^\circ - \theta) = \frac{OM'}{OP'} = -\frac{x}{r} = -\cos \theta$$

$$\text{and, } \tan(180^\circ - \theta) = \frac{P'M'}{OM'} = \frac{y}{-x} = -\tan \theta$$

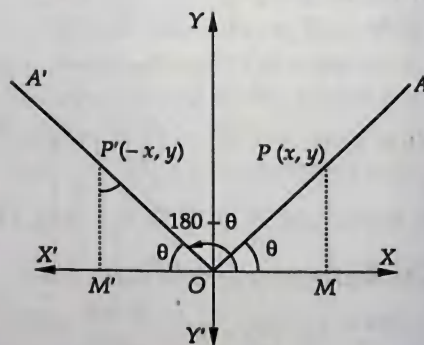


Fig. 5.11

Similarly, we obtain

$$\operatorname{cosec}(180^\circ - \theta) = \operatorname{cosec} \theta, \sec(180^\circ - \theta) = -\sec \theta \text{ and } \cot(180^\circ - \theta) = -\cot \theta$$

5.6.5 TRIGONOMETRICAL RATIOS OF $(180^\circ + \theta)$ IN TERMS OF THAT OF θ

Let a revolving ray OA starting from its initial position OX , trace out an angle $\angle XO A = \theta$. Let another revolving line OA' starting from its position OX , trace out an angle $\angle XO A' = 180^\circ + \theta$. Let P and P' be points on OA and OA' respectively such that $OP = OP' = r$. Draw perpendiculars PM and $P'M'$ from P and P' respectively on OX and OX' . Clearly, triangles OPM and $OP'M'$ are congruent.

$$\therefore OM' = OM = x, PM = P'M' = y$$

Hence, the coordinates of P' are $(-x, -y)$.

$$\text{Clearly, } \sin(180^\circ + \theta) = \frac{P'M'}{OP'} = \frac{-y}{r} = -\frac{y}{r} = -\sin \theta$$

$$\cos(180^\circ + \theta) = \frac{OM'}{OP'} = \frac{-x}{r} = -\frac{x}{r} = -\cos \theta$$

$$\tan(180^\circ + \theta) = \frac{P'M'}{OM'} = \frac{-y}{-x} = \frac{y}{x} = \tan \theta$$

Similarly, we obtain

$$\operatorname{cosec}(180^\circ + \theta) = -\operatorname{cosec} \theta, \sec(180^\circ + \theta) = -\sec \theta \text{ and } \cot(180^\circ + \theta) = \cot \theta.$$

5.6.6 TRIGONOMETRIC RATIOS OF $(360^\circ - \theta)$ IN TERMS OF THAT OF θ

The terminal sides of co-terminal angles coincide, so their trigonometrical ratios are same.

Clearly, $360^\circ - \theta$ and $-\theta$ are co-terminal angles.

$$\therefore \sin(360^\circ - \theta) = \sin(-\theta) = -\sin \theta, \cos(360^\circ - \theta) = \cos(-\theta) = \cos \theta,$$

$$\text{and, } \tan(360^\circ - \theta) = \tan(-\theta) = -\tan \theta$$

Similarly, we have

$$\operatorname{cosec}(360^\circ - \theta) = -\operatorname{cosec} \theta, \sec(360^\circ - \theta) = \sec \theta \text{ and } \cot(360^\circ - \theta) = -\cot \theta.$$

We know that the terminal sides of co-terminal angles always coincide and θ and $360^\circ + \theta$ are co-terminal angles. Therefore,

$$\sin(360^\circ + \theta) = \sin \theta, \cos(360^\circ + \theta) = \cos \theta, \tan(360^\circ + \theta) = \tan \theta,$$

$$\sec(360^\circ + \theta) = \sec \theta, \operatorname{cosec}(360^\circ + \theta) = \operatorname{cosec} \theta \text{ and } \cot(360^\circ + \theta) = \cot \theta$$

In fact, for any positive integer n , angle $(360^\circ \times n + \theta)$ is coterminal to angle θ . Therefore, for any positive integer n , we have

$$\sin(360^\circ \times n + \theta) = \sin \theta, \cos(360^\circ \times n + \theta) = \cos \theta, \tan(360^\circ \times n + \theta) = \tan \theta,$$

$$\operatorname{cosec}(360^\circ \times n + \theta) = \operatorname{cosec} \theta, \sec(360^\circ \times n + \theta) = \sec \theta \text{ and } \cot(360^\circ \times n + \theta) = \cot \theta.$$

If θ is in radians, then the above results may be written as

$$\sin(2n\pi + \theta) = \sin \theta, \cos(2n\pi + \theta) = \cos \theta, \tan(2n\pi + \theta) = \tan \theta,$$

$$\operatorname{cosec}(2n\pi + \theta) = \operatorname{cosec} \theta, \sec(2n\pi + \theta) = \sec \theta \text{ and } \cot(2n\pi + \theta) = \cot \theta$$

5.7 PERIODIC FUNCTION

A function $f(x)$ is said to be a periodic function if there exists a real number $T > 0$ such that $f(x + T) = f(x)$ for all x .

If T is the smallest positive real number such that $f(x + T) = f(x)$ for all x , then T is called the fundamental period of $f(x)$.

Since $\sin(2n\pi + \theta) = \sin \theta$, $\cos(2n\pi + \theta) = \cos \theta$ for all values of θ and $n \in \mathbb{N}$.

Therefore, sine and cosine functions are periodic functions.

We find that 2π is the smallest positive real number such that $\sin(2\pi + \theta) = \sin \theta$, and $\cos(2\pi + \theta) = \cos \theta$ for all values of θ .

So, sine and cosine functions are periodic with period 2π .

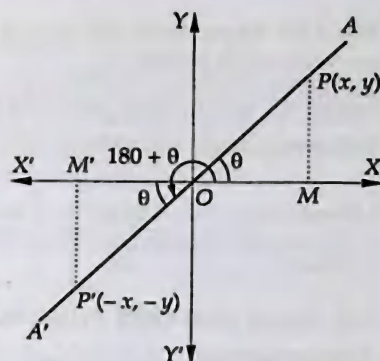


Fig. 5.12

We also know that $\tan(\pi + \theta) = \tan \theta$ and $\cot(\pi + \theta) = \cot \theta$. Therefore, $\tan \theta$ and $\cot \theta$ are periodic with period π .

Similarly, $\operatorname{cosec} \theta$ and $\sec \theta$ are periodic functions with period 2π .

Following table provides periods of all trigonometric functions as a ready reference.

Function:	Sine	Cosine	Tangent	Cosecant	Secant	Cotangent
Period:	2π	2π	π	2π	2π	π

5.8 EVEN AND ODD FUNCTIONS

EVEN FUNCTION A function $f(x)$ is said to be an even function, if $f(-x) = f(x)$ for all x in its domain.

ODD FUNCTION A function $f(x)$ is an odd function, if $f(-x) = -f(x)$ for all x in its domain.

We have seen that $\sin(-\theta) = -\sin \theta$, $\cos(-\theta) = \cos \theta$ and $\tan(-\theta) = -\tan \theta$.

Therefore, $\sin \theta$ and $\tan \theta$ and their reciprocals $\operatorname{cosec} \theta$ and $\cot \theta$ are odd functions whereas $\cos \theta$ and its reciprocal $\sec \theta$ are even functions.

In earlier classes, we have learnt about the values of trigonometrical ratios for 0° , 30° , 45° , 60° and 90° . The values of trigonometric functions for these angles are same as that of trigonometrical ratios studied in earlier classes. In fact, the value of a trigonometrical ratio for any angle is same as the value of trigonometrical function for the same angle. Thus, we have the following table:

Angle Trigonometric Function	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined	0	not defined	0
cosec	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	not defined	-1	not defined
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined	-1	not defined	1
cot	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	not defined	0	not defined

In order to find the values of trigonometrical functions for any angle in terms of those of positive acute angle, we may follow the following algorithm:

ALGORITHM

STEP I See whether the given angle α is positive or negative if it is negative, make it positive by using the following :

$$\sin(-\theta) = -\sin \theta, \cos(-\theta) = \cos \theta, \tan(-\theta) = -\tan \theta \text{ etc.}$$

STEP II Express the positive angle α obtained in step I in the form $\alpha = 90^\circ \times n \pm \theta$, where θ is an acute angle.

STEP III Determine the quadrant in which the terminal side of the angle α lies.

STEP IV Determine the sign of the given trigonometrical function in the quadrant obtained in step III.

STEP V If n in step II is an odd integer, then $\sin \alpha = \pm \cos \theta$, $\cos \alpha = \pm \sin \theta$, $\tan \alpha = \pm \cot \theta$, $\sec \alpha = \pm \operatorname{cosec} \theta$ and $\operatorname{cosec} \theta = \pm \sec \theta$. The sign on RHS will be the sign obtained in step IV.

If n in step II is an even integer, then $\sin \alpha = \pm \sin \theta$, $\cos \alpha = \pm \cos \theta$, $\tan \alpha = \pm \tan \theta$, $\sec \alpha = \pm \sec \theta$ and $\operatorname{cosec} \alpha = \pm \operatorname{cosec} \theta$. The sign on RHS is the sign obtained in step IV.

Following examples will illustrate the above algorithm.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find the values of the following trigonometric ratios:

- (i) $\sin 315^\circ$ (ii) $\cos 210^\circ$ (iii) $\cos (-480^\circ)$ (iv) $\sin (-1125^\circ)$

SOLUTION (i) Clearly,

$$\sin 315^\circ = \sin (90^\circ \times 3 + 45^\circ)$$

Since 315° lies in the IVth quadrant in which sine function is negative and 3 is an odd integer.

$$\therefore \sin 315^\circ = \sin (90^\circ \times 3 + 45^\circ) = -\cos 45^\circ = -\frac{1}{\sqrt{2}}$$

(ii) Clearly,

$$\cos 210^\circ = \cos (90^\circ \times 2 + 30^\circ).$$

Since 210° is in the III quadrant in which cosine function is negative. Also the multiple of 90° is even.

$$\therefore \cos 210^\circ = \cos (90^\circ \times 2 + 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

(iii) Clearly,

$$\cos (-480^\circ) = \cos 480^\circ = \cos (90^\circ \times 5 + 30^\circ)$$

Since 480° is in the II quadrant in which cosine function is negative. Also the multiple of 90° is odd.

$$\therefore \cos (-480^\circ) = \cos 480^\circ = \cos (90^\circ \times 5 + 30^\circ) = -\sin 30^\circ = -\frac{1}{2}$$

(iv) We have,

$$\sin (-1125^\circ) = -\sin 1125^\circ = -\sin (90^\circ \times 12 + 45^\circ)$$

Clearly, 1125° lies in the first quadrant. The multiple of 90° in this expression is even.

$$\therefore \sin (-1125^\circ) = -\sin (90^\circ \times 12 + 45^\circ) = -\sin 45^\circ = -\frac{1}{\sqrt{2}}$$

EXAMPLE 2 Find the values of the following trigonometric ratios.

- (i) $\operatorname{cosec} 390^\circ$ (ii) $\cot 570^\circ$ (iii) $\tan 480^\circ$ (iv) $\cos 270^\circ$
 (v) $\tan \frac{19\pi}{3}$ (vi) $\sin \left(-\frac{11\pi}{3} \right)$ (vii) $\cot \left(-\frac{15\pi}{4} \right)$

SOLUTION (i) We have,

$$390^\circ = 90^\circ \times 4 + 30^\circ$$

Clearly, 390° is in I quadrant and the multiple of 90° is even.

$$\therefore \operatorname{cosec} 390^\circ = \operatorname{cosec} (90^\circ \times 4 + 30^\circ) = \operatorname{cosec} 30^\circ = 2$$

(ii) We have,

$$570^\circ = 90^\circ \times 6 + 30^\circ$$

Clearly, 570° is in the IIIrd quadrant and the multiple of 90° is even.

$$\therefore \cot 570^\circ = \cot (90^\circ \times 6 + 30^\circ) = \cot 30^\circ = \sqrt{3}$$

(iii) We have,

$$480^\circ = 90^\circ \times 5 + 30^\circ$$

Clearly, 480° is in the second quadrant and the multiple of 90° is odd.

$$\therefore \tan 480^\circ = \tan (90^\circ \times 5 + 30^\circ) = -\cot 30^\circ = -\sqrt{3}$$

(iv) We have,

$$270^\circ = 90^\circ \times 3 + 0^\circ$$

Clearly, 270° is in the negative direction of y -axis i.e. on the boundary line of II and III quadrant. Also, the multiple of 90° is an odd integer.

$$\therefore \cos 270^\circ = \cos (90^\circ \times 3 + 0^\circ) = \pm \sin 0^\circ = 0$$

(v) We have,

$$\frac{19\pi}{3} = \left(\frac{19}{3} \times 180 \right)^\circ = 1140^\circ = 90^\circ \times 12 + 60^\circ$$

Clearly, this angle lies in first quadrant.

$$\therefore \tan \frac{19\pi}{3} = \tan (90^\circ \times 12 + 60^\circ) = \tan 60^\circ = \sqrt{3}$$

(vi) We have,

$$\frac{11\pi}{3} = \left(\frac{11 \times 180^\circ}{3} \right) = 660^\circ = 90^\circ \times 7 + 30^\circ$$

$$\therefore \sin \left(-\frac{11\pi}{3} \right) = -\sin \frac{11\pi}{3} = -\sin (90^\circ \times 7 + 30^\circ) = -(-\cos 30^\circ) = -\left(-\frac{\sqrt{3}}{2} \right) = \frac{\sqrt{3}}{2}$$

(vii) We have,

$$\frac{15\pi}{4} = \left(\frac{15 \times 180}{4} \right)^\circ = 675^\circ = 90^\circ \times 7 + 45^\circ$$

$$\therefore \cot \left(-\frac{15\pi}{4} \right) = -\cot \left(\frac{15\pi}{4} \right) = -\cot (90^\circ \times 7 + 45^\circ) = -(-\cot 45^\circ) = -(-1) = 1.$$

EXAMPLE 3 Prove that: $\cos 510^\circ \cos 330^\circ + \sin 390^\circ \cos 120^\circ = -1$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos 510^\circ \cos 330^\circ + \sin 390^\circ \cos 120^\circ \\ &= \cos (90^\circ \times 5 + 60^\circ) \cos (90^\circ \times 3 + 60^\circ) + \sin (90^\circ \times 4 + 30^\circ) \cos (90^\circ \times 1 + 30^\circ) \\ &= (-\sin 60^\circ) (\sin 60^\circ) + (\sin 30^\circ) (-\sin 30^\circ) \\ &= -\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \left(\frac{1}{2} \right) \left(-\frac{1}{2} \right) = -\frac{3}{4} - \frac{1}{4} = -1 = \text{RHS} \end{aligned}$$

EXAMPLE 4 Prove that: $\sin (-420^\circ) (\cos 390^\circ) + \cos (-660^\circ) (\sin 330^\circ) = -1$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \sin (-420^\circ) (\cos 390^\circ) + \cos (-660^\circ) (\sin 330^\circ) \\ &= -\sin 420^\circ \cos 390^\circ + \cos 660^\circ \sin 330^\circ \quad [\because \sin (-\theta) = -\sin \theta, \cos (-\theta) = \cos \theta] \\ &= -\sin (90^\circ \times 4 + 60^\circ) \cos (90^\circ \times 4 + 30^\circ) + \cos (90^\circ \times 7 + 30^\circ) \sin (90^\circ \times 3 + 60^\circ) \\ &= -(\sin 60^\circ) (\cos 30^\circ) + (\sin 30^\circ) (-\cos 60^\circ) = -\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \left(-\frac{1}{2} \right) = -1 = \text{RHS} \end{aligned}$$

EXAMPLE 5 Prove that:

$$(i) \sin^2 \frac{\pi}{6} + \cos^2 \frac{\pi}{3} - \tan^2 \frac{\pi}{4} = -\frac{1}{2} \quad (ii) 2 \sin^2 \frac{\pi}{6} + \operatorname{cosec}^2 \frac{7\pi}{6} \cos^2 \frac{\pi}{3} = \frac{3}{2}$$

$$(iii) \cot^2 \frac{\pi}{6} + \operatorname{cosec} \frac{5\pi}{6} + 3 \tan^2 \frac{\pi}{6} = 6 \quad (iv) 2 \sin^2 \frac{3\pi}{4} + 2 \cos^2 \frac{\pi}{4} + 2 \sec^2 \frac{\pi}{3} = 10$$

SOLUTION (i) $\text{LHS} = \sin^2 \frac{\pi}{6} + \cos^2 \frac{\pi}{3} - \tan^2 \frac{\pi}{4}$
 $= \left(\sin \frac{\pi}{6}\right)^2 + \left(\cos \frac{\pi}{3}\right)^2 - \left(\tan \frac{\pi}{4}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 - (1)^2 = \frac{1}{4} + \frac{1}{4} - 1 = -\frac{1}{2}$

(ii) $\text{LHS} = 2 \sin^2 \frac{\pi}{6} + \operatorname{cosec}^2 \frac{7\pi}{6} \cos^2 \frac{\pi}{3}$
 $= 2 \left(\sin \frac{\pi}{6}\right)^2 + \left(\operatorname{cosec} \frac{7\pi}{6}\right)^2 \left(\cos \frac{\pi}{3}\right)^2$
 $= 2 \left(\sin \frac{\pi}{6}\right)^2 + \left\{\operatorname{cosec} \left(\pi + \frac{\pi}{6}\right)\right\}^2 \left(\cos \frac{\pi}{3}\right)^2$
 $= 2 \left(\sin \frac{\pi}{6}\right)^2 + \left\{-\operatorname{cosec} \frac{\pi}{6}\right\}^2 \left(\cos \frac{\pi}{3}\right)^2 \quad [\because \operatorname{cosec}(\pi + \theta) = -\operatorname{cosec} \theta]$
 $= 2 \left(\frac{1}{2}\right)^2 + (-2)^2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{2} + 1 = \frac{3}{2} = \text{RHS}$

(iii) $\text{LHS} = \cot^2 \frac{\pi}{6} + \operatorname{cosec} \frac{5\pi}{6} + 3 \tan^2 \frac{\pi}{6}$
 $= \left(\cot \frac{\pi}{6}\right)^2 + \operatorname{cosec} \left(\pi - \frac{\pi}{6}\right) + 3 \left(\tan \frac{\pi}{6}\right)^2 = (\sqrt{3})^2 + 2 + 3 \left(\frac{1}{\sqrt{3}}\right)^2 = 3 + 2 + 1 = 6 = \text{RHS}$

(iv) $\text{LHS} = 2 \sin^2 \frac{3\pi}{4} + 2 \cos^2 \frac{\pi}{4} + 2 \sec^2 \frac{\pi}{3}$
 $= 2 \left(\sin \frac{3\pi}{4}\right)^2 + 2 \left(\cos \frac{\pi}{4}\right)^2 + 2 \left(\sec \frac{\pi}{3}\right)^2$
 $= 2 \left(\sin \frac{\pi}{4}\right)^2 + 2 \left(\cos \frac{\pi}{4}\right)^2 + 2 \left(\sec \frac{\pi}{3}\right)^2 \quad \left[\because \sin \frac{3\pi}{4} = \sin \left(\pi - \frac{\pi}{4}\right) = \sin \frac{\pi}{4}\right]$
 $= 2 \left(\frac{1}{\sqrt{2}}\right)^2 + 2 \left(\frac{1}{\sqrt{2}}\right)^2 + 2 (2)^2 = 1 + 1 + 8 = 10 = \text{RHS}$

EXAMPLE 6 ✓ Prove that: $\frac{\cos(90^\circ + \theta) \sec(-\theta) \tan(180^\circ - \theta)}{\sec(360^\circ - \theta) \sin(180^\circ + \theta) \cot(90^\circ - \theta)} = -1$.

SOLUTION We have,

$$\text{LHS} = \frac{\cos(90^\circ + \theta) \sec(-\theta) \tan(180^\circ - \theta)}{\sec(360^\circ - \theta) \sin(180^\circ + \theta) \cot(90^\circ - \theta)} = \frac{(-\sin \theta) (\sec \theta) (-\tan \theta)}{(\sec \theta) (-\sin \theta) (\tan \theta)} = -1 = \text{RHS}$$

EXAMPLE 7 ✓ Prove that :

(i) $\frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos\left(\frac{\pi}{2} + x\right)} = \cot^2 x$

(ii) $\cos\left(\frac{3\pi}{2} + x\right) \cos(2\pi + x) \left\{\cot\left(\frac{3\pi}{2} - x\right) + \cot(2\pi + x)\right\} = 1$

SOLUTION (i) $\text{LHS} = \frac{\cos(\pi + x) \cos(-x)}{\sin(\pi - x) \cos\left(\frac{\pi}{2} + x\right)} = \frac{(-\cos x) \times (\cos x)}{(\sin x) (-\sin x)} = \frac{-\cos^2 x}{-\sin^2 x} = \cot^2 x = \text{RHS}$

(ii) $\text{LHS} = \cos\left(\frac{3\pi}{2} + x\right) \cos(2\pi + x) \left\{\cot\left(\frac{3\pi}{2} - x\right) + \cot(2\pi + x)\right\}$

$$\begin{aligned}
 &= (\sin x)(\cos x)(\tan x + \cot x) \\
 &= \sin x \cos x \left\{ \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right\} \\
 &= \sin x \cos x \left\{ \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \right\} = \sin x \cos x \times \frac{1}{\sin x \cos x} = 1
 \end{aligned}
 \quad \left[\begin{array}{l} \because \cos\left(\frac{3\pi}{2} + x\right) = \sin x, \cos(2\pi + x) = \cos x \\ \cot\left(\frac{3\pi}{2} - x\right) = \tan x \text{ and } \cot(2\pi + x) = \cot x \end{array} \right]$$

EXAMPLE 8 If A, B, C, D are angles of a cyclic quadrilateral, prove that
 $\cos A + \cos B + \cos C + \cos D = 0$.

SOLUTION We know that the opposite angles of a cyclic quadrilateral are supplementary i.e.
 $A + C = \pi$ and $B + D = \pi$.

$$\therefore A = \pi - C \text{ and } B = \pi - D$$

$$\Rightarrow \cos A = \cos(\pi - C) = -\cos C \text{ and } \cos B = \cos(\pi - D) = -\cos D$$

$$\therefore \cos A + \cos B + \cos C + \cos D = -\cos C - \cos D + \cos C + \cos D = 0$$

EXAMPLE 9 In any quadrilateral $ABCD$, prove that

$$(i) \sin(A + B) + \sin(C + D) = 0$$

$$(ii) \cos(A + B) = \cos(C + D)$$

SOLUTION We have,

$$(i) A + B + C + D = 2\pi$$

$$\Rightarrow A + B = 2\pi - (C + D)$$

$$\Rightarrow \sin(A + B) = \sin\{2\pi - (C + D)\}$$

$$\Rightarrow \sin(A + B) = -\sin(C + D)$$

$$\Rightarrow \sin(A + B) + \sin(C + D) = 0$$

$$[\because \sin(2\pi - \theta) = -\sin \theta]$$

(ii) We have,

$$A + B + C + D = 2\pi$$

$$\Rightarrow A + B = 2\pi - (C + D)$$

$$\Rightarrow \cos(A + B) = \cos\{2\pi - (C + D)\}$$

$$\Rightarrow \cos(A + B) = \cos(C + D)$$

$$[\because \cos(2\pi - \theta) = \cos \theta]$$

LEVEL-2

EXAMPLE 10 Find the value of the expression

$$3 \left\{ \sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta) \right\} - 2 \left\{ \sin^6\left(\frac{\pi}{2} + \theta\right) + \sin^6(5\pi - \theta) \right\}$$

SOLUTION The given expression is

[NCERT EXEMPLAR]

$$3 \left\{ \sin^4\left(\frac{3\pi}{2} - \theta\right) + \sin^4(3\pi + \theta) \right\} - 2 \left\{ \sin^6\left(\frac{\pi}{2} + \theta\right) + \sin^6(5\pi - \theta) \right\}$$

$$= 3 \left\{ (-\cos \theta)^4 + (-\sin \theta)^4 \right\} - 2 \left\{ (\cos \theta)^6 + (\sin \theta)^6 \right\}$$

$$= 3(\cos^4 \theta + \sin^4 \theta) - 2(\cos^6 \theta + \sin^6 \theta)$$

$$= 3 \left\{ (\cos^2 \theta + \sin^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta \right\}$$

$$- 2 \left\{ (\cos^2 \theta + \sin^2 \theta)^3 - 3 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta) \right\}$$

$$= 3(1 - 2 \sin^2 \theta \cos^2 \theta) - 2(1 - 3 \cos^2 \theta \sin^2 \theta)$$

$$= 3 - 6 \sin^2 \theta \cos^2 \theta - 2 + 6 \sin^2 \theta \cos^2 \theta = 1.$$

LEVEL-1

1. Find the values of the following trigonometric ratios:

- (i) $\sin \frac{5\pi}{3}$ (ii) $\sin 3060^\circ$ (iii) $\tan \frac{11\pi}{6}$ (iv) $\cos (-1125^\circ)$
 (v) $\tan 315^\circ$ (vi) $\sin 510^\circ$ (vii) $\cos 570^\circ$ (viii) $\sin (-330^\circ)$
 (ix) $\operatorname{cosec} (-1200^\circ)$ (x) $\tan (-585^\circ)$ (xi) $\cos 855^\circ$ (xii) $\sin 1845^\circ$
 (xiii) $\cos 1755^\circ$ (xiv) $\sin 4530^\circ$

2. Prove that:

- (i) $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ = 0$
 (ii) $\sin \frac{8\pi}{3} \cos \frac{23\pi}{6} + \cos \frac{13\pi}{3} \sin \frac{35\pi}{6} = \frac{1}{2}$
 (iii) $\cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ + \cos 300^\circ = \frac{1}{2}$
 (iv) $\tan (-225^\circ) \cot (-405^\circ) - \tan (-765^\circ) \cot (675^\circ) = 0$
 (v) $\cos 570^\circ \sin 510^\circ + \sin (-330^\circ) \cos (-390^\circ) = 0$
 (vi) $\tan \frac{11\pi}{3} - 2 \sin \frac{4\pi}{6} - \frac{3}{4} \operatorname{cosec}^2 \frac{\pi}{4} + 4 \cos^2 \frac{17\pi}{6} = \frac{3-4\sqrt{3}}{2}$
 (vii) $3 \sin \frac{\pi}{6} \sec \frac{\pi}{3} - 4 \sin \frac{5\pi}{6} \cot \frac{\pi}{4} = 1$

3. Prove that:

- (i) $\frac{\cos (2\pi + \theta) \operatorname{cosec} (2\pi + \theta) \tan (\pi/2 + \theta)}{\sec (\pi/2 + \theta) \cos \theta \cot (\pi + \theta)} = 1$
 (ii) $\frac{\operatorname{cosec} (90^\circ + \theta) + \cot (450^\circ + \theta)}{\operatorname{cosec} (90^\circ - \theta) + \tan (180^\circ - \theta)} + \frac{\tan (180^\circ + \theta) + \sec (180^\circ - \theta)}{\tan (360^\circ + \theta) - \sec (-\theta)} = 2$
 (iii) $\frac{\sin (180^\circ + \theta) \cos (90^\circ + \theta) \tan (270^\circ - \theta) \cot (360^\circ - \theta)}{\sin (360^\circ - \theta) \cos (360^\circ + \theta) \operatorname{cosec} (-\theta) \sin (270^\circ + \theta)} = 1$
 (iv) $\left\{ 1 + \cot \theta - \sec \left(\frac{\pi}{2} + \theta \right) \right\} \left\{ 1 + \cot \theta + \sec \left(\frac{\pi}{2} + \theta \right) \right\} = 2 \cot \theta$
 (v) $\frac{\tan (90^\circ - \theta) \sec (180^\circ - \theta) \sin (-\theta)}{\sin (180^\circ + \theta) \cot (360^\circ - \theta) \operatorname{cosec} (90^\circ - \theta)} = 1$

4. Prove that : $\sin^2 \frac{\pi}{18} + \sin^2 \frac{\pi}{9} + \sin^2 \frac{7\pi}{18} + \sin^2 \frac{4\pi}{9} = 2$

5. Prove that : $\sec \left(\frac{3\pi}{2} - \theta \right) \sec \left(\theta - \frac{5\pi}{2} \right) + \tan \left(\frac{5\pi}{2} + \theta \right) \tan \left(\theta - \frac{3\pi}{2} \right) = -1$.

6. In a ΔABC , prove that :

- (i) $\cos (A + B) + \cos C = 0$ (ii) $\cos \left(\frac{A+B}{2} \right) = \sin \frac{C}{2}$ (iii) $\tan \frac{A+B}{2} = \cot \frac{C}{2}$

7. If A, B, C, D be the angles of a cyclic quadrilateral, taken in order, prove that :

$$\cos (180^\circ - A) + \cos (180^\circ + B) + \cos (180^\circ + C) - \sin (90^\circ + D) = 0$$

8. Find x from the following equations :

(i) $\operatorname{cosec} (90^\circ + \theta) + x \cos \theta \cot (90^\circ + \theta) = \sin (90^\circ + \theta)$

(ii) $x \cot (90^\circ + \theta) + \tan (90^\circ + \theta) \sin \theta + \operatorname{cosec} (90^\circ + \theta) = 0$

9. Prove that:

(i) $\tan 720^\circ - \cos 270^\circ - \sin 150^\circ \cos 120^\circ = \frac{1}{4}$

- (ii) $\sin 780^\circ \sin 480^\circ + \cos 120^\circ \sin 150^\circ = \frac{1}{2}$
 (iii) $\sin 780^\circ \sin 120^\circ + \cos 240^\circ \sin 390^\circ = \frac{1}{2}$
 (iv) $\sin 600^\circ \cos 390^\circ + \cos 480^\circ \sin 150^\circ = -1$
 (v) $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ = 0$

ANSWERS

1. (i) $-\frac{\sqrt{3}}{2}$ (ii) 0 (iii) $-\frac{1}{\sqrt{3}}$ (iv) $\frac{1}{\sqrt{2}}$ (v) -1 (vi) $\frac{1}{2}$
 (vii) $-\frac{\sqrt{3}}{2}$ (viii) $\frac{1}{2}$ (ix) $-\frac{2}{\sqrt{3}}$ (x) -1 (xi) $-\frac{1}{\sqrt{2}}$ (xii) $\frac{1}{\sqrt{2}}$
 (xiii) $1/\sqrt{2}$ (xiv) $-1/2$ 8. (i) $\tan \theta$ (ii) $\sin \theta$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- Write the maximum and minimum values of $\cos (\cos x)$.
- Write the maximum and minimum values of $\sin (\sin x)$.
- Write the maximum value of $\sin (\cos x)$.
- If $\sin x = \cos^2 x$, then write the value of $\cos^2 x (1 + \cos^2 x)$.
- If $\sin x + \operatorname{cosec} x = 2$, then write the value of $\sin^n x + \operatorname{cosec}^n x$.
- If $\sin x + \sin^2 x = 1$, then write the value of $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x$.
- If $\sin x + \sin^2 x = 1$, then write the value of $\cos^8 x + 2 \cos^6 x + \cos^4 x$.
- If $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3$, then write the value of $\cos \theta_1 + \cos \theta_2 + \cos \theta_3$.
- Write the value of $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ$.
- A circular wire of radius 15 cm is cut and bent so as to lie along the circumference of a loop of radius 120 cm. Write the measure of the angle subtended by it at the centre of the loop.
- Write the value of $2 (\sin^6 \theta + \cos^6 \theta) - 3 (\sin^4 \theta + \cos^4 \theta) + 1$.
- Write the value of $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 180^\circ$.
- If $\cot (\alpha + \beta) = 0$, then write the value of $\sin (\alpha + 2\beta)$.
- If $\tan A + \cot A = 4$, then write the value of $\tan^4 A + \cot^4 A$.
- Write the least value of $\cos^2 \theta + \sec^2 \theta$.
- If $x = \sin^{14} \theta + \cos^{20} \theta$, then write the smallest interval in which the value of x lie.
- If $3 \sin \theta + 5 \cos \theta = 5$, then write the value of $5 \sin \theta - 3 \cos \theta$.

ANSWERS

1. 1, $\cos 1$ 2. $\sin 1, -\sin 1$ 3. $\sin 1$ 4. 1 5. 2 6. 1
 7. 1 8. 0 9. 0 10. 45° 11. 0 12. 0 13. $\sin \alpha$ or $\cos \beta$
 14. 194 15. 2 16. $(0, 1]$ 17. 3 or -3

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

1. If $\tan \theta = x - \frac{1}{4x}$, then $\sec \theta - \tan \theta$ is equal to

- (a) $-2x, \frac{1}{2x}$ (b) $-\frac{1}{2x}, 2x$ (c) $2x$ (d) $2x, \frac{1}{2x}$
2. If $\sec \theta = x + \frac{1}{4x}$, then $\sec \theta + \tan \theta =$
 (a) $x, \frac{1}{x}$ (b) $2x, \frac{1}{2x}$ (c) $-2x, \frac{1}{2x}$ (d) $-\frac{1}{x}, x$
3. If $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$, then $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}}$ is equal to
 (a) $\sec \theta - \tan \theta$ (b) $\sec \theta + \tan \theta$ (c) $\tan \theta - \sec \theta$ (d) none of these
4. If $\pi < \theta < 2\pi$, then $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$ is equal to
 (a) $\operatorname{cosec} \theta + \cot \theta$ (b) $\operatorname{cosec} \theta - \cot \theta$ (c) $-\operatorname{cosec} \theta + \cot \theta$ (d) $-\operatorname{cosec} \theta - \cot \theta$
5. If $0 < \theta < \frac{\pi}{2}$, and if $\frac{y+1}{1-y} = \sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$, then y is equal to
 (a) $\cot \frac{\theta}{2}$ (b) $\tan \frac{\theta}{2}$ (c) $\cot \frac{\theta}{2} + \tan \frac{\theta}{2}$ (d) $\cot \frac{\theta}{2} - \tan \frac{\theta}{2}$
6. If $\frac{\pi}{2} < \theta < \pi$, then $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} + \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$ is equal to
 (a) $2 \sec \theta$ (b) $-2 \sec \theta$ (c) $\sec \theta$ (d) $-\sec \theta$
7. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$, then $x^2 + y^2 + z^2$ is independent of
 (a) θ, ϕ (b) r, θ (c) r, ϕ (d) r
8. If $\tan \theta + \sec \theta = \sqrt{3}$, $0 < \theta < \pi$, then θ is equal to
 (a) $\frac{5\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$
9. If $\tan \theta = -\frac{1}{\sqrt{5}}$ and θ lies in the IV quadrant, then the value of $\cos \theta$ is
 (a) $\frac{\sqrt{5}}{\sqrt{6}}$ (b) $\frac{2}{\sqrt{6}}$ (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{6}}$
10. If $\frac{3\pi}{4} < \alpha < \pi$, then $\sqrt{2 \cot \alpha + \frac{1}{\sin^2 \alpha}}$ is equal to
 (a) $1 - \cot \alpha$ (b) $1 + \cot \alpha$ (c) $-1 + \cot \alpha$ (d) $-1 - \cot \alpha$
11. $\sin^6 A + \cos^6 A + 3 \sin^2 A \cos^2 A =$
 (a) 0 (b) 1 (c) 2 (d) 3
12. If $\operatorname{cosec} \theta - \cot \theta = \frac{1}{2}$, $0 < \theta < \frac{\pi}{2}$, then $\cos \theta$ is equal to
 (a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $-\frac{3}{5}$ (d) $-\frac{5}{3}$
13. If $\operatorname{cosec} \theta + \cot \theta = \frac{11}{2}$, then $\tan \theta =$
 (a) $\frac{21}{22}$ (b) $\frac{15}{16}$ (c) $\frac{44}{117}$ (d) $\frac{117}{44}$
14. $\sec^2 \theta = \frac{4xy}{(x+y)^2}$ is true if and only if
 (a) $x + y \neq 0$ (b) $x = y, x \neq 0$ (c) $x = y$ (d) $x \neq 0, y \neq 0$

15. If θ is an acute angle and $\tan \theta = \frac{1}{\sqrt{7}}$, then the value of $\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta}$ is
 (a) $3/4$ (b) $1/2$ (c) 2 (d) $5/4$
16. The value of $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 85^\circ + \sin^2 90^\circ$ is
 (a) 7 (b) 8 (c) 9.5 (d) 10
17. $\sin^2 \frac{\pi}{18} + \sin^2 \frac{\pi}{9} + \sin^2 \frac{7\pi}{18} + \sin^2 \frac{4\pi}{9} =$
 (a) 1 (b) 4 (c) 2 (d) 0
18. If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ is equal to
 (a) 110 (b) 191 (c) 80 (d) 194
19. If $x \sin 45^\circ \cos^2 60^\circ = \frac{\tan^2 60^\circ \operatorname{cosec} 30^\circ}{\sec 45^\circ \cot^2 30^\circ}$, then $x =$
 (a) 2 (b) 4 (c) 8 (d) 16
20. If $\operatorname{cosec} \theta - \cot \theta = \frac{1}{2}$, $0 < \theta < \frac{\pi}{2}$, then $\cos \theta$ is equal to
 (a) $-3/5$ (b) $-5/3$ (c) $5/3$ (d) $3/5$
21. If $\operatorname{cosec} A + \cot A = \frac{11}{2}$, then $\tan A =$
 (a) $21/22$ (b) $15/16$ (c) $44/117$ (d) $117/43$
22. If $\tan \theta + \sec \theta = e^x$, then $\cos \theta$ equals
 (a) $\frac{e^x + e^{-x}}{2}$ (b) $\frac{2}{e^x + e^{-x}}$ (c) $\frac{e^x - e^{-x}}{2}$ (d) $\frac{e^x - e^{-x}}{e^x + e^{-x}}$
23. If $\sec \theta + \tan \theta = k$, $\cos \theta =$
 (a) $\frac{k^2 + 1}{2k}$ (b) $\frac{2k}{k^2 + 1}$ (c) $\frac{k}{k^2 + 1}$ (d) $\frac{k}{k^2 - 1}$
24. If $f(x) = \cos^2 x + \sec^2 x$, then
 (a) $f(x) < 1$ (b) $f(x) = 1$ (c) $2 < f(x) < 1$ (d) $f(x) \geq 2$
25. Which of the following is incorrect?
 (a) $\sin \theta = -1/5$ (b) $\cos \theta = 1$ (c) $\sec \theta = 1/2$ (d) $\tan \theta = 20$
26. The value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 179^\circ$ is
 (a) $1/\sqrt{2}$ (b) 0 (c) 1 (d) -1
27. The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$ is
 (a) 0 (b) 1 (c) $1/2$ (d) not defined
28. Which of the following is correct?
 (a) $\sin 1^\circ > \sin 1$ (b) $\sin 1^\circ < \sin 1$ (c) $\sin 1^\circ = \sin 1$ (d) $\sin 1^\circ = \frac{\pi}{180} \sin 1$
29. If A lies in second quadrant and $3 \tan A + 4 = 0$, then the value of $2 \cot A - 5 \cos A + \sin A$ is equal to
 (a) $-53/10$ (b) $23/10$ (c) $37/10$ (d) $7/10$

ANSWERS

- | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (c) | 4. (d) | 5. (b) | 6. (b) | 7. (a) | 8. (c) |
| 9. (a) | 10. (d) | 11. (b) | 12. (b) | 13. (c) | 14. (b) | 15. (a) | 16. (c) |
| 17. (c) | 18. (d) | 19. (c) | 20. (d) | 21. (c) | 22. (b) | 23. (b) | 24. (d) |
| 25. (c) | 26. (b) | 27. (b) | 28. (b) | 29. (b) | | | |

SUMMARY

1. Following are some of the fundamental trigonometric identities:

$$(i) \sin \theta = \frac{1}{\operatorname{cosec} \theta} \text{ or, } \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$(ii) \cos \theta = \frac{1}{\sec \theta} \text{ or, } \sec \theta = \frac{1}{\cos \theta} \quad (iii) \cot \theta = \frac{1}{\tan \theta} \text{ or, } \tan \theta = \frac{1}{\cot \theta}$$

$$(iv) \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ or, } \cot \theta = \frac{\cos \theta}{\sin \theta} \quad (v) \sin^2 \theta + \cos^2 \theta = 1$$

$$(vi) 1 + \tan^2 \theta = \sec^2 \theta \text{ or, } \sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$(vii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta \text{ or, } \operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

2. (i) $\sin(-\theta) = -\sin \theta$ or, $\operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$

(ii) $\cos(-\theta) = \cos \theta$ or, $\sec(-\theta) = \sec \theta$

(iii) $\tan(-\theta) = -\tan \theta$ or, $\cot(-\theta) = -\cot \theta$

(iv) $\sin(90^\circ - \theta) = \cos \theta$, $\cos(90^\circ - \theta) = \sin \theta$

$\tan(90^\circ - \theta) = \cot \theta$, $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$

$\operatorname{cosec}(90^\circ - \theta) = \sec \theta$, $\cot(90^\circ - \theta) = \tan \theta$

(v) $\sin(90^\circ + \theta) = \cos \theta$, $\cos(90^\circ + \theta) = -\sin \theta$

$\tan(90^\circ + \theta) = -\cot \theta$, $\cot(90^\circ + \theta) = -\tan \theta$

$\sec(90^\circ + \theta) = -\operatorname{cosec} \theta$, $\operatorname{cosec}(90^\circ + \theta) = \sec \theta$

(vi) $\sin(180^\circ - \theta) = \sin \theta$, $\cos(180^\circ - \theta) = -\cos \theta$

$\tan(180^\circ - \theta) = -\tan \theta$, $\cot(180^\circ - \theta) = -\cot \theta$

$\sec(180^\circ - \theta) = -\sec \theta$, $\operatorname{cosec}(180^\circ - \theta) = \operatorname{cosec} \theta$

(vii) $\sin(270^\circ - \theta) = -\cos \theta$, $\cos(270^\circ - \theta) = -\sin \theta$

$\tan(270^\circ - \theta) = \cot \theta$, $\cot(270^\circ - \theta) = \tan \theta$

$\operatorname{cosec}(270^\circ - \theta) = -\sec \theta$, $\sec(270^\circ - \theta) = -\operatorname{cosec} \theta$

(viii) $\sin(270^\circ + \theta) = -\cos \theta$, $\cos(270^\circ + \theta) = \sin \theta$

$\tan(270^\circ + \theta) = -\cot \theta$, $\cot(270^\circ + \theta) = -\tan \theta$

$\operatorname{cosec}(270^\circ + \theta) = -\sec \theta$, $\sec(270^\circ + \theta) = \operatorname{cosec} \theta$

(ix) $\sin(360^\circ - \theta) = -\sin \theta$, $\cos(360^\circ - \theta) = \cos \theta$

$\tan(360^\circ - \theta) = -\tan \theta$, $\operatorname{cosec}(360^\circ - \theta) = -\operatorname{cosec} \theta$

$\sec(360^\circ - \theta) = \sec \theta$, $\cot(360^\circ - \theta) = -\cot \theta$

(x) Sine and Cosine functions and their reciprocals i.e. Cosecant and Secant functions are periodic functions with period 2π . Tangent and Cotangent functions are periodic with period π .

(xi) *Odd functions*

sine, tangent

cotangent, cosecant

Even functions

cosine, secant